GROWING THE DUAL REGULARIZATION, NOT THE PENALTY: A FIXED-PENALTY SINGLE-LOOP AUGMENTED LAGRANGIAN METHOD FOR NONCONVEX COMPOSITE OPTIMIZATION

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Abstract. We study a class of structured nonconvex constrained optimization problems with nonsmooth composite terms. We develop a dual-damping linearized augmented Lagrangian method and analyze its approximate KKT residual guarantees.

Key words. nonsmooth composite optimization, augmented Lagrangian method, dual-regularization, iteration complexity

AMS subject classifications. 65K05, 90C26, 90C30, 90C46

1. Introduction.

In this paper, we study the structured nonsmooth nonconvex composite problem

$$(\mathbf{P}) \qquad \min_{\boldsymbol{x}\in\mathbb{R}^n} \quad F(\boldsymbol{x}) := f(\boldsymbol{x}) + h(\mathcal{A}(\boldsymbol{x})),$$

where $f\colon \mathbb{R}^n \to \mathbb{R}$ is a possibly nonconvex function with a Lipschitz continuous gradient, $\mathcal{A}\colon \mathbb{R}^n \to \mathbb{R}^m$ is a smooth and possibly nonlinear mapping, and $h\colon \mathbb{R}^m \to \mathbb{R}\cup\{+\infty\}$ is a proper closed convex function whose proximal mapping is easy to evaluate; the complete assumptions are stated in Section 2.1. The two components play complementary roles: f is usually a smooth loss or data-fidelity term, whereas h is a nonsmooth convex function that imposes robustness, sparsity, or other structure on the transformed variable $\mathcal{A}(\boldsymbol{x})$ [4, 23, 26]. Several models of practical interest take this form.

1.1. Motivating Examples.

Robust phase retrieval [6]. Given measurement vectors $\mathbf{a}_i \in \mathbb{R}^n$ and observations $b_i \approx \langle \mathbf{a}_i, \boldsymbol{x}^\star\rangle^2$, a fraction of which may be grossly corrupted, one solves

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} \ \frac{1}{m}\sum_{i=1}^{m} \big|\langle \mathbf{a}_i, \boldsymbol{x}\rangle^2 - b_i\big|.$$

This is Problem (**P**) with $f \equiv 0$, $h = \frac{1}{m}\|\cdot\|_1$, and $\mathcal{A}(\boldsymbol{x})_i = \langle \mathbf{a}_i, \boldsymbol{x}\rangle^2 - b_i$; the ℓ_1 loss makes the model robust to outliers.

Robust low-rank matrix recovery [4]. Given a linear measurement map $\mathcal{M}\colon \mathbb{R}^{d_1\times d_2} \to \mathbb{R}^m$ and observations $\mathbf{b}$ containing outliers, a rank-r matrix is recovered in factorized form via

$$\min_{\mathbf{U}\in\mathbb{R}^{d_1\times r},\, \mathbf{V}\in\mathbb{R}^{d_2\times r}} \ \frac{1}{m}\big\|\mathcal{M}(\mathbf{U}\mathbf{V}^\top) - \mathbf{b}\big\|_1.$$

Here $\boldsymbol{x} = (\mathbf{U}, \mathbf{V})$, $h = \frac{1}{m}\|\cdot\|_1$, and $\mathcal{A}(\boldsymbol{x}) = \mathcal{M}(\mathbf{U}\mathbf{V}^\top) - \mathbf{b}$ is bilinear in $(\mathbf{U}, \mathbf{V})$, so the model again has the form Problem (**P**).

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Minimizing a pointwise maximum [36]. For smooth losses $f_1, \dots, f_m$, such as the losses on m data subgroups, one solves

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} \max_{1\le i\le m} f_i(\boldsymbol{x}).$$

This is Problem (**P**) with $h(u) = \max_{1\le i\le m} u_i$ and $\mathcal{A}(\boldsymbol{x}) = (f_1(\boldsymbol{x}), \dots, f_m(\boldsymbol{x}))$; a smooth regularizer, if present, enters through f, and the proximal mapping of h reduces to a projection onto the unit simplex.

1.2. Related Work. Introducing the auxiliary variable $\boldsymbol{y} = \mathcal{A}(\boldsymbol{x})$ rewrites Problem (**P**) as the equality-constrained problem

$$\min_{\boldsymbol{x},\boldsymbol{y}} f(\boldsymbol{x}) + h(\boldsymbol{y}) \quad \text{s.t.} \quad \mathcal{A}(\boldsymbol{x}) = \boldsymbol{y}, \tag{1.1}$$

which brings it within the scope of penalty methods and augmented Lagrangian methods (ALMs). For general nonlinearly constrained nonconvex problems, both families control feasibility through the penalty parameter. Quadratic penalty methods increase the penalty without bound [13, 17]. ALMs were originally introduced to avoid such unbounded penalties [11, 27], and in convex programming any fixed penalty suffices [29]; in the nonconvex setting, however, inexact ALMs still increase the penalty across outer iterations to obtain complexity guarantees [31, 16], and proximal ALM, which keeps the penalty fixed within a run, chooses its value according to the target accuracy [33]. Single-loop methods have attracted growing attention because they avoid solving subproblems to high accuracy, yet they rely on the same mechanism: single-loop quadratic penalty schemes increase the penalty [1], and linearized ALMs use small dual steps oraccuracy *eps*-dependent parameters [32]. A large penalty inflates the smoothness constant of the primal update, which forces small step sizes and makes the primal subproblems increasingly ill-conditioned.

The split formulation (1.1) also carries favorable structure. Its constraint Jacobian $[J_{\mathcal{A}}(\boldsymbol{x}), -I]$ has full row rank at every $\boldsymbol{x}$, so constraint qualifications hold automatically, and its objective separates into a smooth part in $\boldsymbol{x}$ and a prox-friendly part in $\boldsymbol{y}$. Nonconvex ADMM and related Lagrangian-based splitting methods exploit exactly this structure [2, 5, 9, 8]. The structure, however, does not remove the need for a large penalty. For nonlinear $\mathcal{A}$, adaptive Lagrangian schemes require penalty levels proportional to $1/\epsilon$ [9], and linearized perturbed ALMs use accuracy-dependent penalty and perturbation parameters [8]. The nonconvex–linear minimax route attains $\mathcal{O}(\epsilon^{-3})$ [25, 10, 34], but its dual regularization acts as an effective penalty that grows as the target accuracy decreases. Fixed-penalty guarantees are available for smooth nonlinear equality constraints under a uniform LICQ [7, 19], but they do not cover the nonsmooth term $h(\boldsymbol{y})$ in (1.1). It therefore remains open whether a single-loop ALM can keep every penalty-type parameter independent of both the accuracy and the target accuracy while retaining an explicit first-order update for $h \circ \mathcal{A}$.

From another perspective, Problem (**P**) also admits a minimax reformulation. In particular, eliminating $\boldsymbol{y}$ from the Lagrangian of (1.1) gives

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} \max_{\boldsymbol{\lambda}\in\mathbb{R}^m} \mathcal{L}(\boldsymbol{x},\boldsymbol{\lambda}) = \Phi(\boldsymbol{x},\boldsymbol{\lambda}) - h^*(\boldsymbol{\lambda}), \qquad \Phi(\boldsymbol{x},\boldsymbol{\lambda}) := f(\boldsymbol{x}) + \langle \boldsymbol{\lambda}, \mathcal{A}(\boldsymbol{x})\rangle, \tag{1.2}$$

where $\operatorname{dom} h^* \subseteq \mathbb{B}(\mathbf{0}, L_h)$ by Assumption 2. This is a nonconvex–linear instance of the smooth nonconvex–concave (NC-C) minimax problem. Since $\boldsymbol{\lambda}$ is exactly the multiplier of $\mathcal{A}(\boldsymbol{x}) = \boldsymbol{y}$, KKT points of (1.1) and game-stationary (GS) points of (1.2) agree up to constants, and ALMs can be read as single-loop minimax methods. For large-scale problems, we focus on single-loop first-order methods, which update each variable by one explicit step per iteration without an inner solver. The basic method

of this kind for (1.2) is gradient descent–ascent (GDA). It converges slowly because regularity fails on both sides: Φ is only linear in $\boldsymbol{\lambda}$, so the best response $\boldsymbol{\lambda}^\star(\boldsymbol{x})$ is in general set-valued, and Φ is nonconvex in $\boldsymbol{x}$, so $\boldsymbol{x}^\star(\boldsymbol{\lambda})$ need not be unique or stable. Even with two timescales, GDA needs $\mathcal{O}(\epsilon^{-6})$ iterations to reach an ϵ-GS point [18].

Two techniques improve on this rate, each restoring regularity on one side. The first is primal smoothing, which adds $\frac{1}{2\gamma}\|\boldsymbol{x}-\boldsymbol{z}\|^2$ with a fixed γ and moves the center slowly by $\boldsymbol{z}^{t+1} = \boldsymbol{z}^t+\beta(\boldsymbol{x}^{t+1}-\boldsymbol{z}^t)$. The target accuracy enters only through β: with $\beta = \Theta(\epsilon^2)$, the resulting Smoothed GDA finds an ϵ-GS point in $\mathcal{O}(\epsilon^{-4})$ iterations. For the pointwise maximum of finitely many smooth functions, i.e., (1.2) with $h(\boldsymbol{u}) = \max_i u_i$, strict complementarity permits a constant β and improves the bound to $\mathcal{O}(\epsilon^{-2})$ [36].

The second technique is dual regularization, which subtracts $\frac{1}{2s}\|\boldsymbol{\lambda}-\boldsymbol{\mu}\|^2$ from the dual objective and makes it strongly concave. In our reduced augmented-Lagrangian formulation, this regularization has an effective-penalty interpretation:

$$\max_{\boldsymbol{\lambda}} \mathcal{P}_s(\boldsymbol{x},\boldsymbol{\lambda};\boldsymbol{z},\boldsymbol{\mu}) = \mathcal{D}_{\rho+s}(\boldsymbol{x},\boldsymbol{\mu}) + \frac{1}{2\gamma}\|\boldsymbol{x}-\boldsymbol{z}\|^2, \tag{1.3}$$

where $\mathcal{P}_s$ is defined in (2.5) and $\mathcal{D}_{\rho+s}$ is the reduced augmented Lagrangian (2.3) with penalty $\rho+s$; see Lemma 2.10. A zero center $\boldsymbol{\mu} = \mathbf{0}$ recovers a quadratic penalty with coefficient $\rho+s$, which can leave a nonzero feasibility residual when s is fixed. Updating $\boldsymbol{\mu}$ instead provides a multiplier-update mechanism that may recover feasibility while keeping $\rho+s$ fixed; its convergence and complexity depend on the coupled updates.

With the dual center fixed at $\boldsymbol{\mu} = \mathbf{0}$, Perturbed GDA attains an $\mathcal{O}(\epsilon^{-4})$ complexity for finding an ϵ-GS point [21, 35], and Li et al. establish a matching algorithm-specific lower bound [14]. Methods exploiting the NC-L structure attain $\mathcal{O}(\epsilon^{-3})$ [25, 10, 34]. These guarantees rely on accuracy-dependent regularization parameters, corresponding to an effective penalty that increases as ϵ decreases. To improve the iteration complexity, Li et al. combine primal smoothing with zero-centered dual perturbation to develop Perturbed Smoothed GDA, which yields a genius improvement in complexity order, reducing the GS complexity of Perturbed GDA from $\mathcal{O}(\epsilon^{-4})$ to $\mathcal{O}(\epsilon^{-3})$ with a matching lower bound [14]. However, the dual regularization coefficient still satisfies $1/s = \Theta(\epsilon)$.

With a moving dual center $\boldsymbol{\mu}$ and fixed primal and dual regularization coefficients, DS-GDA achieves $\mathcal{O}(\epsilon^{-4})$ complexity for nonconvex–concave problems. Faster rates are available under a uniform one-sided KŁ condition [39]. Without primal smoothing, PPALA reports $\widetilde{\mathcal{O}}(\epsilon^{-2})$ complexity with fixed parameters for functionally constrained optimization [12]. However, its feasibility argument relies on asymptotic reasoning, and a corresponding finite-iteration feasibility bound remains to be verified.

In all guarantees above that reach $\mathcal{O}(\epsilon^{-3})$, some parameter grows with $1/\epsilon$: either the penalty, as in penalty methods and GDPA, or the dual regularization, as in Perturbed Smoothed GDA. Practical ALM implementations, in contrast, usually keep the penalty fixed. Growing the penalty is costly for a single-loop method, because the Lipschitz constant of $\nabla_{\boldsymbol{x}}\mathcal{D}_\rho$ increases with ρ and forces the primal stepsize to shrink. This leads to the following question.

(Q) *To reach $\mathcal{O}(\epsilon^{-3})$, must a single-loop linearized ALM let some parameter grow with $1/\epsilon$? If so, must this growth be carried by the penalty parameter?*

We answer the first part affirmatively and the second part negatively. By Lemma 2.10, the relevant quantity is the effective penalty $\rho+s$. For every choice of the primal and dual center stepsizes, SPLALM requires $\Omega(\epsilon^{-4})$ iterations whenever $\rho+s \le c_1/\epsilon$ (Theorem 3.1). Conversely, with ρ fixed and only s_k increasing, SPLALM reaches an ϵ-KKT point in $\mathcal{O}(\epsilon^{-3})$ iterations with primal and dual stepsizes independent of ϵ (Theorem 6.1). The growth required by the lower bound can thus be carried entirely by the dual regularization, while the penalty stays fixed. Table 1 summarizes representative complexity

Algorithms	Regularization and penalty parameters				Complexity	
	Penalty parameter	Primal smoothing $\frac{1}{2\gamma}\|\boldsymbol{x}-\boldsymbol{z}\|^2$	Dual regularization $-\frac{1}{2s}\|\boldsymbol{\lambda}-\boldsymbol{\mu}\|^2$		Upper bound	Lower bound
			Center $\boldsymbol{\mu}$	Parameter s		
GDA-based Methods						
Smoothed GDA [37, 14]	—	$\gamma=\Theta(1)$	—	—	$\mathcal{O}(\epsilon^{-4})$	$\Omega(\epsilon^{-4})$
Perturbed GDA [21, 35, 14]	—	—	$\boldsymbol{\mu}\equiv\mathbf{0}$	$\Theta(\epsilon^{-1})$	$\mathcal{O}(\epsilon^{-4})$	$\Omega(\epsilon^{-4})$
Perturbed Smoothed GDA [14]	—	$\gamma=\Theta(1)$	$\boldsymbol{\mu}\equiv\mathbf{0}$	$\Theta(\epsilon^{-1})$	$\mathcal{O}(\epsilon^{-3})$	$\Omega(\epsilon^{-3})$
RADA [34]	—	—	$\boldsymbol{\mu}\not\equiv\mathbf{0}$	$(r_\epsilon+\beta_k)^{-1}$	$\mathcal{O}(\epsilon^{-3})$	—
DS-GDA [39]	—	$\gamma=\Theta(1)$	$\boldsymbol{\mu}\not\equiv\mathbf{0}$	$\Theta(1)$	$\underline{\mathcal{O}(\epsilon^{-4})}$	—
Penalty and augmented Lagrangian-based methods						
Quadratic penalty [1]	$\rho_k=\Theta((k+1)^{1/4})$	—	$\boldsymbol{\mu}\equiv\mathbf{0}$	$\Theta((k+1)^{1/4})$	$\widetilde{\mathcal{O}}(\epsilon^{-4})$	—
GDPA [20]	$\rho_k=\Theta(\epsilon^{-1})$	—	$\boldsymbol{\mu}\equiv\mathbf{0}$	$\Theta(\epsilon^{-1})$	$\mathcal{O}(\epsilon^{-3})$	—
PPALA [12]	$\rho=\Theta(1)$	—	$\boldsymbol{\mu}\not\equiv\mathbf{0}$	$\Theta(1)$	$\widetilde{\mathcal{O}}(\epsilon^{-2})$	—
SPLALM *Theorem 3.1*	$\rho>0$	$\gamma=\Theta(1)$	any	$\rho+s\le c_1\epsilon^{-1}$	?	$\Omega(\epsilon^{-4})$
SPLALM *Theorem 6.1*	$\rho=\Theta(1)$	$\gamma=\Theta(1)$	$\boldsymbol{\mu}\not\equiv\mathbf{0}$	$\Theta((k+1)^{1/3})$	$\mathcal{O}(\epsilon^{-3})$	?

Table 1: Representative single-loop methods, grouped by their original formulation. The constrained rows concern nonlinear constraints. Upper bounds display the dependence on ϵ, suppressing problem constants and bounded initial gaps. The regularization columns specify parameters and centers, not gradient stepsizes; primal smoothing uses a separate moving center. "—" denotes an absent regularizer or an unreported lower bound; question marks denote results to be established. Constant parameters are independent of ϵ and k. The rates use each method's stated stationarity criterion and assumptions; lower bounds are algorithm-specific.

guarantees.

Our contributions are as follows.

1. **A single-loop ALM family and its effective penalty.** We study SPLALM, which performs alternating gradient descent–ascent steps with moving primal and dual centers on the reduced augmented Lagrangian $\mathcal{D}_\rho$ with a fixed penalty $\rho > 0$ (Definition 2.6). The family contains the update structures of quadratic-penalty, doubly smoothed GDA, smoothed GDA, and linearized ALM methods acting on $\mathcal{D}_\rho$ (Table 2). Every member has the effective penalty $\rho + s$ (Lemma 2.10), and its ϵ-KKT certificates are exactly ϵ-game-stationary points of the Fenchel saddle reformulation (Proposition 2.4).
2. **A lower bound uniform in the center stepsizes.** For every pair of center stepsizes $(\alpha, \beta) \in [0,1]^2$ and every (ρ, s) with $\epsilon(\rho + s) \le c_1$, we construct an instance with an affine inner mapping on which the KKT certificates generated by SPLALM require $\Omega(\epsilon^{-4})$ iterations (Theorem 3.1). The initialization is fixed at zero, and only the instance depends on the parameters. A strictly convex perturbation of the outer function converts this bound into a lower bound of the same order for game stationarity of the output pairs $(\boldsymbol{x}^{k+1}, \boldsymbol{q}^k)$. Hence moving the centers cannot compensate for an effective penalty $\rho + s = o(\epsilon^{-1})$.
3. **An upper bound without increasing the penalty.** With ρ, γ, and the primal and dual stepsizes fixed, and with $s_k = \Theta((k+1)^{1/3})$ and diminishing center stepsizes, SPLALM finds an ϵ-KKT point in $\mathcal{O}(\epsilon^{-3})$ iterations (Theorem 6.1). Together with the lower bound, this identifies ϵ^{-1} as the necessary scale of the effective penalty and shows that the dual regularization, not the penalty, can carry it.

Organization. Section 2 introduces the problem class, the KKT and game-stationarity criteria, the SPLALM family, and the complexity measure, and shows that every member of the family has the effective penalty $\rho + s$. Section 3 states the lower bound for constant center stepsizes, and Section 4 proves it. Section 5 extends the lower bound to time-varying center stepsizes. Section 6 proves the $\mathcal{O}(\epsilon^{-3})$ upper bound with a fixed penalty.

2. Preliminaries. Notation. We use $\langle \cdot, \cdot \rangle$ and $\| \cdot \|$ for the Euclidean inner product and norm; for a matrix, $\| \cdot \|$ is the spectral norm. For $\mathcal{A} = (A_1, \dots, A_m)^\top$, the Jacobian $J_{\mathcal{A}}(\boldsymbol{x}) \in \mathbb{R}^{m \times n}$ has ith row $\nabla A_i(\boldsymbol{x})^\top$. For a nonempty closed convex set C, $\operatorname{dist}(\boldsymbol{u}, C) := \min_{\boldsymbol{v} \in C} \|\boldsymbol{u} - \boldsymbol{v}\|$, and $\Pi_C(\boldsymbol{u})$ is the Euclidean projection of $\boldsymbol{u}$ onto C.

For a proper closed convex function $h : \mathbb{R}^m \to \mathbb{R} \cup \{+\infty\}$, its subdifferential, convex conjugate, and proximal mapping are

$$\partial h(\boldsymbol{y}) := \{\boldsymbol{v} : \ h(\boldsymbol{w}) \ge h(\boldsymbol{y}) + \langle \boldsymbol{v}, \boldsymbol{w} - \boldsymbol{y} \rangle \text{ for all } \boldsymbol{w}\}, \qquad h^*(\boldsymbol{q}) := \sup_{\boldsymbol{y} \in \mathbb{R}^m} \{\langle \boldsymbol{q}, \boldsymbol{y} \rangle - h(\boldsymbol{y})\},$$

$$\operatorname{prox}_{th}(\boldsymbol{u}) := \operatorname*{argmin}_{\boldsymbol{y} \in \mathbb{R}^m} \Big\{ h(\boldsymbol{y}) + \frac{1}{2t} \|\boldsymbol{y} - \boldsymbol{u}\|^2 \Big\}, \qquad t > 0; \tag{2.1}$$

see [28]. We write $\operatorname{dom} \partial h^* := \{\boldsymbol{q} : \partial h^*(\boldsymbol{q}) \neq \varnothing\}$.

2.1. Assumptions. We impose the following assumptions.

Assumption 1 (Smooth components). There exist constants $L_f, G_f, L_{\mathcal{A}}, G_{\mathcal{A}} \ge 0$ such that, for all $\boldsymbol{x}, \boldsymbol{x}' \in \mathbb{R}^n$,

$$\|\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{x}')\| \le L_f \|\boldsymbol{x} - \boldsymbol{x}'\|, \quad \|\nabla f(\boldsymbol{x})\| \le G_f,$$
$$\|J_{\mathcal{A}}(\boldsymbol{x}) - J_{\mathcal{A}}(\boldsymbol{x}')\| \le L_{\mathcal{A}} \|\boldsymbol{x} - \boldsymbol{x}'\|, \quad \|J_{\mathcal{A}}(\boldsymbol{x})\| \le G_{\mathcal{A}}.$$

Assumption 2 (Lipschitz outer function). The function $h : \mathbb{R}^m \to \mathbb{R}$ is convex and L_h-Lipschitz:

$$|h(\boldsymbol{u}) - h(\boldsymbol{v})| \le L_h \|\boldsymbol{u} - \boldsymbol{v}\|, \qquad \boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^m.$$

2.2. Problem class, stationarity, and oracle. To study (Q), we introduce the following problem class.

DEFINITION 2.1 (Problem class). *Let $\Delta > 0$ and $\mathbf{C} = (L_f, G_f, L_{\mathcal{A}}, G_{\mathcal{A}}, L_h) \in \mathbb{R}^5_{\geq 0}$. We denote by $\mathcal{I}(\mathbf{C}, \Delta)$ the set of all triples $(f, \mathcal{A}, h)$, over all dimensions $n, m \in \mathbb{N}$, such that Assumptions 1 and 2 hold with the constants in $\mathbf{C}$ and $F(\mathbf{0}) - \inf_{\boldsymbol{x}\in\mathbb{R}^n} F(\boldsymbol{x}) \leq \Delta$.*

The constants $\mathbf{C}$ and Δ do not depend on the target accuracy or on n, m, and the gap condition implies $\inf F > -\infty$.

We use the following stationarity measure for Problem (**P**).

DEFINITION 2.2 (ϵ-KKT point). *Given $\epsilon > 0$. A triple $(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{q})$ with $\boldsymbol{q} \in \partial h(\boldsymbol{y})$ is an ϵ-KKT point of (**P**) if*

$$\mathcal{R}_{\mathrm{KKT}}(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{q}) := \max\Big\{ \big\|\nabla f(\boldsymbol{x}) + J_{\mathcal{A}}(\boldsymbol{x})^\top \boldsymbol{q}\big\|,\ \|\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}\| \Big\} \leq \epsilon \tag{2.2}$$

We next relate Definition 2.2 to game stationarity (GS) for the minimax reformulation Problem (1.2) of Problem (**P**). We use the following GS definition from [15, Definition 3.1(b)] and [38, (2.10)–(2.11)].

DEFINITION 2.3 (ϵ-GS point). *Let $\epsilon > 0$. A pair $(\boldsymbol{x}, \boldsymbol{q}) \in \mathbb{R}^n \times \mathrm{dom}\partial h^*$ is an ϵ-game stationary point (ϵ-GS) of Problem (1.2) if*

$$\mathcal{G}(\boldsymbol{x}, \boldsymbol{q}) := \max\Big\{ \big\|\nabla f(\boldsymbol{x}) + J_{\mathcal{A}}(\boldsymbol{x})^\top \boldsymbol{q}\big\|,\ \mathrm{dist}\big(\mathcal{A}(\boldsymbol{x}), \partial h^*(\boldsymbol{q})\big) \Big\} \leq \epsilon.$$

We now relate ϵ-KKT points of Problem (**P**) to ϵ-GS points of its Fenchel saddle reformulation Problem (1.2).

PROPOSITION 2.4 (KKT points of Problem (**P**) and GS points of Problem (1.2)). *Suppose that Assumption 2 holds, and let $(\boldsymbol{x}, \boldsymbol{q}) \in \mathbb{R}^n \times \mathrm{dom}\partial h^*$. Then the following measurement relationship holds:*

$$\mathcal{G}(\boldsymbol{x}, \boldsymbol{q}) = \min_{\boldsymbol{y}\in\partial h^*(\boldsymbol{q})} \mathcal{R}_{\mathrm{KKT}}(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{q}),$$

with minimum attained at $\hat{\boldsymbol{y}} = \Pi_{\partial h^*(\boldsymbol{q})}\big(\mathcal{A}(\boldsymbol{x})\big)$. Consequently, for every $\epsilon > 0$:

*(a) if $(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{q})$ is an ϵ-KKT point of Problem (**P**), then $(\boldsymbol{x}, \boldsymbol{q})$ is an ϵ-GS point of Problem (1.2);*

*(b) if $(\boldsymbol{x}, \boldsymbol{q})$ is an ϵ-GS point of Problem (1.2), then $(\boldsymbol{x}, \hat{\boldsymbol{y}}, \boldsymbol{q})$ is an ϵ-KKT point of Problem (**P**).*

Proof. Since h is proper, closed, and convex, $\boldsymbol{q} \in \partial h(\boldsymbol{y})$ if and only if $\boldsymbol{y} \in \partial h^*(\boldsymbol{q})$ [28, Theorem 23.5]. The first term of $\mathcal{R}_{\mathrm{KKT}}$ does not depend on $\boldsymbol{y}$. The set $\partial h^*(\boldsymbol{q})$ is nonempty, closed, and convex, so $\min_{\boldsymbol{y}\in\partial h^*(\boldsymbol{q})} \|\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}\| = \mathrm{dist}(\mathcal{A}(\boldsymbol{x}), \partial h^*(\boldsymbol{q}))$, attained at $\hat{\boldsymbol{y}}$. Statements (a) and (b) follow. ∎

We next introduce the following deterministic first-order oracle model.

DEFINITION 2.5 (Oracle model). *We assume access to a deterministic first-order oracle*

$$\mathcal{O}_{f,\mathcal{A}}(\boldsymbol{x}) := \big(f(\boldsymbol{x}), \nabla f(\boldsymbol{x}), \mathcal{A}(\boldsymbol{x}), J_{\mathcal{A}}(\boldsymbol{x})\big),$$

and an exact proximal oracle

$$\mathcal{O}_h(\boldsymbol{v}, t) := \mathrm{prox}_{th}(\boldsymbol{v}), \qquad t > 0.$$

Each call to either oracle counts as one oracle evaluation. In particular, the cost of solving the proximal subproblem is represented by one exact proximal-oracle call.

2.3. Algorithm family and worst-case iteration complexity. For $\rho > 0$, the reduced augmented Lagrangian of (1.1) is

$$\mathcal{D}_\rho(\boldsymbol{x}, \boldsymbol{\lambda}) := \min_{\boldsymbol{y} \in \mathbb{R}^m} \Big\{ f(\boldsymbol{x}) + h(\boldsymbol{y}) + \langle \boldsymbol{\lambda}, \mathcal{A}(\boldsymbol{x}) - \boldsymbol{y} \rangle + \frac{\rho}{2} \|\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}\|^2 \Big\}. \tag{2.3}$$

Its minimizer and the associated multiplier are

$$\boldsymbol{y}(\boldsymbol{x}, \boldsymbol{\lambda}) := \operatorname{prox}_{h/\rho}\Big(\mathcal{A}(\boldsymbol{x}) + \frac{\boldsymbol{\lambda}}{\rho}\Big), \qquad \boldsymbol{q}(\boldsymbol{x}, \boldsymbol{\lambda}) := \boldsymbol{\lambda} + \rho\big[\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}(\boldsymbol{x}, \boldsymbol{\lambda})\big], \tag{2.4}$$

and $\boldsymbol{q}(\boldsymbol{x}, \boldsymbol{\lambda}) \in \partial h(\boldsymbol{y}(\boldsymbol{x}, \boldsymbol{\lambda}))$. We study the smoothed perturbed linearized augmented Lagrangian method (SPLALM), which performs alternating gradient descent–ascent steps on the smoothed perturbed potential function

$$\mathcal{P}_s(\boldsymbol{x}, \boldsymbol{\lambda}; \boldsymbol{z}, \boldsymbol{\mu}) := \mathcal{D}_\rho(\boldsymbol{x}, \boldsymbol{\lambda}) + \frac{1}{2\gamma}\|\boldsymbol{x} - \boldsymbol{z}\|^2 - \frac{1}{2s}\|\boldsymbol{\lambda} - \boldsymbol{\mu}\|^2. \tag{2.5}$$

The proximal term smooths the primal problem, the term $-\frac{1}{2s}\|\boldsymbol{\lambda} - \boldsymbol{\mu}\|^2$ perturbs the dual problem, and $\boldsymbol{z} \in \mathbb{R}^n$ and $\boldsymbol{\mu} \in \mathbb{R}^m$ are the primal and dual centers.

DEFINITION 2.6 (SPLALM). *Let $\theta = (\rho, s, \gamma, \eta, \tau)$ satisfy $\rho > 0$, $0 < \eta \le \gamma < \infty$, and $0 < \tau \le s < \infty$, and let $(\alpha, \beta) \in [0,1]^2$. The method* $\mathrm{SPLALM}_{\theta,\alpha,\beta}$ *starts from $(\boldsymbol{x}^0, \boldsymbol{\lambda}^0, \boldsymbol{z}^0, \boldsymbol{\mu}^0) = \mathbf{0}$ and, for $k \ge 0$, sets*

$$\begin{aligned}
\boldsymbol{x}^{k+1} &= \boldsymbol{x}^k - \eta\, \nabla_{\boldsymbol{x}} \mathcal{P}_s(\boldsymbol{x}^k, \boldsymbol{\lambda}^k; \boldsymbol{z}^k, \boldsymbol{\mu}^k),\\
\boldsymbol{\lambda}^{k+1} &= \boldsymbol{\lambda}^k + \tau\, \nabla_{\boldsymbol{\lambda}} \mathcal{P}_s(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k; \boldsymbol{z}^k, \boldsymbol{\mu}^k),\\
\boldsymbol{z}^{k+1} &= \boldsymbol{z}^k + \alpha(\boldsymbol{x}^{k+1} - \boldsymbol{z}^k),\\
\boldsymbol{\mu}^{k+1} &= \boldsymbol{\mu}^k + \beta(\boldsymbol{\lambda}^{k+1} - \boldsymbol{\mu}^k).
\end{aligned} \tag{2.6}$$

At iteration k, it outputs $(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}, \boldsymbol{q}^k) := (\boldsymbol{x}^{k+1}, \boldsymbol{y}(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k), \boldsymbol{q}(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k))$.

With first-order information cached between iterations, K iterations and their outputs use $K + 1$ calls to $\mathcal{O}_{f,\mathcal{A}}$ and $2K$ calls to $\mathcal{O}_h$. Thus the total oracle count is $3K + 1$, and iteration and oracle complexity agree up to a factor of three, apart from one initial first-order oracle call.

DEFINITION 2.7 (SPLALM family). *For θ as in Definition 2.6, $\mathcal{C}_\theta := \{\mathrm{SPLALM}_{\theta,\alpha,\beta} : (\alpha, \beta) \in [0,1]^2\}$.*

DEFINITION 2.8 (Worst-case iteration complexity). *Let $\epsilon > 0$ and $I \in \mathcal{I}(\mathbf{C}, \Delta)$, and let $(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}, \boldsymbol{q}^k)$ be the outputs of* $\mathrm{SPLALM}_{\theta,\alpha,\beta}$ *on I. Define*

$$\begin{aligned}
T_\epsilon^{\mathrm{KKT}}(\theta, \alpha, \beta; I) &:= \inf\Big\{ K \ge 1 : \min_{0 \le k < K} \mathcal{R}_{\mathrm{KKT}}(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}, \boldsymbol{q}^k) \le \epsilon \Big\},\\
T_\epsilon^{\mathrm{GS}}(\theta, \alpha, \beta; I) &:= \inf\Big\{ K \ge 1 : \min_{0 \le k < K} \mathcal{G}(\boldsymbol{x}^{k+1}, \boldsymbol{q}^k) \le \epsilon \Big\},
\end{aligned}$$

with $\inf \varnothing = +\infty$. For $\bullet \in \{\mathrm{KKT}, \mathrm{GS}\}$, the worst-case iteration complexity of $\mathcal{C}_\theta$ over $\mathcal{I}(\mathbf{C}, \Delta)$ is

$$T_\epsilon^\bullet\big(\mathcal{C}_\theta, \mathcal{I}(\mathbf{C}, \Delta)\big) := \inf_{(\alpha,\beta) \in [0,1]^2} \sup_{I \in \mathcal{I}(\mathbf{C},\Delta)} T_\epsilon^\bullet(\theta, \alpha, \beta; I).$$

The infimum precedes the supremum, so (α, β) may depend on ϵ, θ, and $(\mathbf{C}, \Delta)$, but not on the instance. At least one of the first K outputs meets the tolerance if and only if $K \geq T_\epsilon^\bullet$; hence the definition covers every rule that returns one of these outputs, including the best-so-far rule. By Proposition 2.4, $\mathcal{G}(\boldsymbol{x}^{k+1}, \boldsymbol{q}^k) \leq \mathcal{R}_{\mathrm{KKT}}(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}, \boldsymbol{q}^k)$ for every k, so $T_\epsilon^{\mathrm{GS}}(\mathcal{C}_\theta, \mathcal{I}) \leq T_\epsilon^{\mathrm{KKT}}(\mathcal{C}_\theta, \mathcal{I})$. An upper bound for the KKT measure therefore yields the same upper bound for the GS measure, and a lower bound for the GS measure yields the same lower bound for the KKT measure.

Remark 2.9 (Family complexity and oracle complexity). Let $\mathcal{A}_{\mathrm{det}}$ be the class of deterministic methods whose iterates are arbitrary maps of all previous responses of $\mathcal{O}_{f,\mathcal{A}}$ and $\mathcal{O}_h$, defined as in [3, Section 2.2] and [24, Definition 2.5] with the oracles of Definition 2.5. Then $\mathcal{C}_\theta \subset \mathcal{A}_{\mathrm{det}}$, and

$$T_\epsilon(\mathcal{A}_{\mathrm{det}}, \mathcal{I}) = \inf_{A \in \mathcal{A}_{\mathrm{det}}} \sup_{I \in \mathcal{I}} T_\epsilon(A; I) \leq \inf_{(\alpha,\beta)\in[0,1]^2} \sup_{I \in \mathcal{I}} T_\epsilon(\theta, \alpha, \beta; I) = T_\epsilon(\mathcal{C}_\theta, \mathcal{I}).$$

(i) (Scope) A lower bound on $T_\epsilon(\mathcal{C}_\theta, \mathcal{I})$ does not bound $T_\epsilon(\mathcal{A}_{\mathrm{det}}, \mathcal{I})$; methods outside $\mathcal{C}_\theta$ may be faster. When h is the support function of a compact convex set, (1.2) is a smooth nonconvex–concave problem, and the $O(\epsilon^{-5/2})$ GS bound of [38] and the $O(\epsilon^{-3})$ OS bound of [24] are attained by methods outside $\mathcal{C}_\theta$.

(ii) (Uniformity) The infimum ranges over all $(\alpha, \beta) \in [0, 1]^2$, including the endpoint cases in Table 2, and Theorem 3.1 holds for all (ρ, s) with $\epsilon(\rho + s) \leq c_1$. A lower bound on $T_\epsilon(\mathcal{C}_\theta, \mathcal{I})$ therefore cannot be removed by tuning the center stepsizes; it is governed by the effective penalty $\rho + s$ of Lemma 2.10.

(iii) (Transfer) Conversely, every lower bound on $T_\epsilon(\mathcal{A}_{\mathrm{det}}, \mathcal{I})$ for the same problem class and measure also holds for $\mathcal{C}_\theta$.

2.4. Effective penalty and special cases. The reduced augmented Lagrangian is a dual proximal smoothing of the saddle function $\mathcal{L}(\boldsymbol{x}, \boldsymbol{q}) = f(\boldsymbol{x}) + \langle \mathcal{A}(\boldsymbol{x}), \boldsymbol{q}\rangle - h^*(\boldsymbol{q})$ in (1.2) [30]. The center of this smoothing is the multiplier, and its strength is the penalty. Adding the dual regularizer of (2.5) keeps this form; it only raises the penalty and moves the center.

LEMMA 2.10 (Effective penalty). *Suppose that Assumption 2 holds. For every $\sigma > 0$,*

$$\mathcal{D}_\sigma(\boldsymbol{x}, \boldsymbol{\lambda}) = \max_{\boldsymbol{q} \in \mathbb{R}^m} \Big\{ \mathcal{L}(\boldsymbol{x}, \boldsymbol{q}) - \frac{1}{2\sigma} \|\boldsymbol{q} - \boldsymbol{\lambda}\|^2 \Big\}, \tag{2.7}$$

and consequently

$$\max_{\boldsymbol{\lambda} \in \mathbb{R}^m} \mathcal{P}_s(\boldsymbol{x}, \boldsymbol{\lambda}; \boldsymbol{z}, \boldsymbol{\mu}) = \mathcal{D}_{\rho+s}(\boldsymbol{x}, \boldsymbol{\mu}) + \frac{1}{2\gamma} \|\boldsymbol{x} - \boldsymbol{z}\|^2. \tag{2.8}$$

Proof. Write $\boldsymbol{u} = \mathcal{A}(\boldsymbol{x})$ and $h(\boldsymbol{y}) = \max_{\boldsymbol{q}}\{\langle \boldsymbol{q}, \boldsymbol{y}\rangle - h^*(\boldsymbol{q})\}$. Since $\mathrm{dom} h^*$ lies in a ball by Assumption 2, Sion's minimax theorem allows the minimum over $\boldsymbol{y}$ and the maximum over $\boldsymbol{q}$ in $\min_{\boldsymbol{y}} \max_{\boldsymbol{q}}\{\langle \boldsymbol{q}-\boldsymbol{\lambda}, \boldsymbol{y}\rangle - h^*(\boldsymbol{q}) + \frac{\sigma}{2}\|\boldsymbol{u}-\boldsymbol{y}\|^2\}$ to be interchanged. The inner minimizer is $\boldsymbol{y} = \boldsymbol{u} - (\boldsymbol{q}-\boldsymbol{\lambda})/\sigma$, which gives (2.7). Applying (2.7) with $\sigma = \rho$ and exchanging the two maximizations,

$$\max_{\boldsymbol{\lambda}} \mathcal{P}_s = \max_{\boldsymbol{q}} \Big\{ \mathcal{L}(\boldsymbol{x}, \boldsymbol{q}) - \min_{\boldsymbol{\lambda}} \Big[\frac{\|\boldsymbol{q} - \boldsymbol{\lambda}\|^2}{2\rho} + \frac{\|\boldsymbol{\lambda} - \boldsymbol{\mu}\|^2}{2s} \Big] \Big\} + \frac{\|\boldsymbol{x} - \boldsymbol{z}\|^2}{2\gamma},$$

and the inner minimum equals $\|\boldsymbol{q} - \boldsymbol{\mu}\|^2/(2(\rho + s))$. Then (2.7) with $\sigma = \rho + s$ gives (2.8). ∎

Lemma 2.10 shows that every member of $\mathcal{C}_\theta$ has the effective penalty $\rho + s$, whatever the center stepsizes are. The center stepsize β only places the dual center between two classical choices, and α decides whether the primal variable is smoothed.

Table 2: Special cases of $\mathcal{C}_\theta$. All methods act on $\mathcal{D}_\rho$ with $\rho > 0$ and have the effective penalty $\rho + s$.

β	Dual center	$\alpha = 1$	$0 < \alpha < 1$
0	$\mathbf{0}$	perturbed GDA [21, 35]	perturbed smoothed GDA [14]
$(0, 1)$	$\sum_{j=1}^{k} \beta(1-\beta)^{k-j}\boldsymbol{\lambda}^j$	—	DS-GDA [39], $r_1 = 1/\gamma$, $r_2 = 1/s$
1	$\boldsymbol{\lambda}^k$	linearized ALM ($\tau = \rho$)	smoothed GDA [37]

Zero center ($\beta = 0$). Then $\boldsymbol{\mu}^k = \mathbf{0}$, and $\mathcal{D}_{\rho+s}(\boldsymbol{x}, \mathbf{0}) = f(\boldsymbol{x}) + \min_{\boldsymbol{y}}\{h(\boldsymbol{y}) + \frac{\rho+s}{2}\|\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}\|^2\}$ is the quadratic penalty function with parameter $\rho + s$. The dual step carries the perturbation $-\boldsymbol{\lambda}^k/s$, as in perturbed GDA [21, 35] and perturbed smoothed GDA [14].

Current multiplier as center ($\beta = 1$). Then $\boldsymbol{\mu}^k = \boldsymbol{\lambda}^k$, the gradient of the dual regularizer vanishes at every dual step, and

$$\boldsymbol{\lambda}^{k+1} = \Big(1 - \frac{\tau}{\rho}\Big)\boldsymbol{\lambda}^k + \frac{\tau}{\rho}\,\boldsymbol{q}^k, \qquad \boldsymbol{q}^k = \operatorname{prox}_{\rho h^*}\big(\boldsymbol{\lambda}^k + \rho\mathcal{A}(\boldsymbol{x}^{k+1})\big).$$

For $\tau = \rho$, this is the multiplier update of the augmented Lagrangian method [11, 27, 29], that is, a proximal ascent step on $\mathcal{L}(\boldsymbol{x}^{k+1}, \cdot)$ by (2.7). The iterates do not depend on s, so this choice also reproduces the case without dual regularizer, $s = \infty$.

Moving center ($0 < \beta < 1$). The dual center is an exponentially weighted average of past multipliers, and the method has the structure of doubly smoothed GDA [39].

Table 2 collects these cases. The identities $\boldsymbol{\mu}^k = \mathbf{0}$, $\boldsymbol{\mu}^k = \boldsymbol{\lambda}^k$, and $\boldsymbol{z}^k = \boldsymbol{x}^k$ for $\beta = 0$, $\beta = 1$, and $\alpha = 1$, respectively, follow by induction from the zero initialization. Every entry refers to the cited update structure applied to $\mathcal{D}_\rho$ with $\rho > 0$; methods acting on $\mathcal{L}$ itself are compared in Section 3.2.

3. Lower Bound for Constant Center Stepsizes.

3.1. Main result. We use the problem class of Definition 2.1, the family $\mathcal{C}_\theta$ of Definition 2.7, and the complexity of Definition 2.8, with the normalized constants $\mathbf{C}_0 := (1, 1, 0, 1, 1)$. Since $L_{\mathcal{A}} = 0$, every instance in $\mathcal{I}(\mathbf{C}_0, \Delta)$ has an affine inner mapping.

THEOREM 3.1 (Lower bound uniform in ρ and s). *Fix $0 < \eta \le \gamma$, $\tau > 0$, and $\Delta > 0$. There exist constants $c_1 = c_1(\gamma, \eta, \tau) > 0$ and $c_2 = c_2(\gamma, \tau, \Delta) > 0$, independent of ρ, s, α, and β, such that the following holds. For every $\epsilon \in (0, c_2)$, every $\rho > 0$ and $s \ge \tau$ with $\epsilon(\rho + s) \le c_1$, and every $(\alpha, \beta) \in [0, 1]^2$, there exists an instance $I \in \mathcal{I}(\mathbf{C}_0, \Delta)$ on which* SPLALM$_{\theta,\alpha,\beta}$ *requires at least*

$$\frac{\Delta\,\epsilon^{-4}}{32768\,\gamma\max\{1, 4\tau, 4\gamma\tau\}}$$

iterations before one of its outputs is an ϵ-KKT point. Equivalently,

$$T_\epsilon^{\mathrm{KKT}}(\mathcal{C}_\theta, \mathcal{I}(\mathbf{C}_0, \Delta)) \ge \Delta\epsilon^{-4}/(32768\,\gamma\max\{1, 4\tau, 4\gamma\tau\})$$

.

TODO: State the GS version T_ϵ^{GS} after the strict-convex perturbation lemma is proved.

Remark 3.2 (Initialization). The tight analyses of GDA-type methods in [22, Theorems 5.1–5.2] and [14, Theorems 3.2 and 3.4] choose the hard function together with the initialization. In Theorem 3.1, the initialization is fixed at zero by Definition 2.6, and only the instance depends on $(\theta, \alpha, \beta, \epsilon)$.

COROLLARY 3.3 (Necessary scale of the effective penalty). *Let $\rho = \rho(\epsilon) > 0$ and $s = s(\epsilon) \geq \tau$. If $\rho + s = o(\epsilon^{-1})$, then Theorem 3.1 applies for all sufficiently small ϵ, and the complexity of $\mathcal{C}_\theta$ is $\Omega(\epsilon^{-4})$. In particular, an $O(\epsilon^{-3})$ guarantee for $\mathcal{C}_\theta$ requires $\rho(\epsilon) + s(\epsilon) > c_1/\epsilon$ for all sufficiently small ϵ.*

3.2. Relation to Smoothed GDA and DS-GDA. $\mathrm{SPLALM}_{\theta,\alpha,\beta}$ in Definition 2.6 uses $\mathrm{prox}_{h/\rho}$, so it requires $\rho > 0$. Smoothed GDA and DS-GDA contain no such term. They nevertheless generate the same iterates as $\mathrm{SPLALM}_{\theta,\alpha,\beta}$ as long as $\boldsymbol{y}(\boldsymbol{x},\boldsymbol{\lambda}) = \mathbf{0}$. To state this, define

$$\mathcal{R}_\rho := \big\{(\boldsymbol{x},\boldsymbol{\lambda}) :\ \boldsymbol{\lambda} + \rho\mathcal{A}(\boldsymbol{x}) \in \partial h(\mathbf{0})\big\}, \qquad \Psi_\rho(\boldsymbol{x},\boldsymbol{\lambda}) := f(\boldsymbol{x}) + \frac{\rho}{2}\|\mathcal{A}(\boldsymbol{x})\|^2 + \langle \boldsymbol{\lambda}, \mathcal{A}(\boldsymbol{x})\rangle.$$

PROPOSITION 3.4 (Reduction on $\mathcal{R}_\rho$). *Let $\rho > 0$ and $(\boldsymbol{x},\boldsymbol{\lambda}) \in \mathcal{R}_\rho$. Then $\boldsymbol{y}(\boldsymbol{x},\boldsymbol{\lambda}) = \mathbf{0}$, $\boldsymbol{q}(\boldsymbol{x},\boldsymbol{\lambda}) = \boldsymbol{\lambda} + \rho\mathcal{A}(\boldsymbol{x})$, and*

$$\nabla f(\boldsymbol{x}) + J_{\mathcal{A}}(\boldsymbol{x})^\top \boldsymbol{q}(\boldsymbol{x},\boldsymbol{\lambda}) = \nabla_{\boldsymbol{x}} \Psi_\rho(\boldsymbol{x},\boldsymbol{\lambda}), \qquad \mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}(\boldsymbol{x},\boldsymbol{\lambda}) = \nabla_{\boldsymbol{\lambda}} \Psi_\rho(\boldsymbol{x},\boldsymbol{\lambda}).$$

Moreover, $\mathcal{D}_\rho(\boldsymbol{x},\boldsymbol{\lambda}) = \Psi_\rho(\boldsymbol{x},\boldsymbol{\lambda}) + h(\mathbf{0})$.

Proof. Set $\boldsymbol{v} := \mathcal{A}(\boldsymbol{x}) + \boldsymbol{\lambda}/\rho$. By (2.4), $\boldsymbol{y}(\boldsymbol{x},\boldsymbol{\lambda})$ is the unique minimizer of $h(\boldsymbol{y}) + \frac{\rho}{2}\|\boldsymbol{y} - \boldsymbol{v}\|^2$, and $\boldsymbol{y}$ is this minimizer if and only if $\rho(\boldsymbol{v} - \boldsymbol{y}) \in \partial h(\boldsymbol{y})$. For $\boldsymbol{y} = \mathbf{0}$ the inclusion reads $\boldsymbol{\lambda} + \rho\mathcal{A}(\boldsymbol{x}) \in \partial h(\mathbf{0})$, which holds because $(\boldsymbol{x},\boldsymbol{\lambda}) \in \mathcal{R}_\rho$. Hence $\boldsymbol{y}(\boldsymbol{x},\boldsymbol{\lambda}) = \mathbf{0}$ and $\boldsymbol{q}(\boldsymbol{x},\boldsymbol{\lambda}) = \boldsymbol{\lambda} + \rho\mathcal{A}(\boldsymbol{x})$. Since $\nabla_{\boldsymbol{x}}\big(\frac{\rho}{2}\|\mathcal{A}(\boldsymbol{x})\|^2\big) = \rho J_{\mathcal{A}}(\boldsymbol{x})^\top \mathcal{A}(\boldsymbol{x})$,

$$\nabla f(\boldsymbol{x}) + J_{\mathcal{A}}(\boldsymbol{x})^\top \big(\boldsymbol{\lambda} + \rho\mathcal{A}(\boldsymbol{x})\big) = \nabla_{\boldsymbol{x}} \Psi_\rho(\boldsymbol{x},\boldsymbol{\lambda}),$$

and $\mathcal{A}(\boldsymbol{x}) - \mathbf{0} = \nabla_{\boldsymbol{\lambda}} \Psi_\rho(\boldsymbol{x},\boldsymbol{\lambda})$. Finally, completing the square in (2.3) gives $\mathcal{D}_\rho(\boldsymbol{x},\boldsymbol{\lambda}) = f(\boldsymbol{x}) + \min_{\boldsymbol{y}}\{h(\boldsymbol{y}) + \frac{\rho}{2}\|\boldsymbol{y} - \boldsymbol{v}\|^2\} - \frac{1}{2\rho}\|\boldsymbol{\lambda}\|^2$, and the minimum equals $h(\mathbf{0}) + \frac{\rho}{2}\|\boldsymbol{v}\|^2$ because its minimizer is $\mathbf{0}$. Hence

$$\mathcal{D}_\rho(\boldsymbol{x},\boldsymbol{\lambda}) = f(\boldsymbol{x}) + h(\mathbf{0}) + \frac{\rho}{2}\|\mathcal{A}(\boldsymbol{x})\|^2 + \langle \boldsymbol{\lambda}, \mathcal{A}(\boldsymbol{x})\rangle + \frac{1}{2\rho}\|\boldsymbol{\lambda}\|^2 - \frac{1}{2\rho}\|\boldsymbol{\lambda}\|^2 = \Psi_\rho(\boldsymbol{x},\boldsymbol{\lambda}) + h(\mathbf{0}). \qquad \square$$

For $\theta = (\rho, s, \gamma, \eta, \tau)$ and $(\alpha,\beta) \in [0,1]^2$, consider the recursion

$$\begin{aligned} \boldsymbol{x}^{k+1} &= \boldsymbol{x}^k - \eta\Big[\nabla_{\boldsymbol{x}} \Psi_\rho(\boldsymbol{x}^k,\boldsymbol{\lambda}^k) + \frac{1}{\gamma}(\boldsymbol{x}^k - \boldsymbol{z}^k)\Big], \\ \boldsymbol{\lambda}^{k+1} &= \boldsymbol{\lambda}^k + \tau\Big[\nabla_{\boldsymbol{\lambda}} \Psi_\rho(\boldsymbol{x}^{k+1},\boldsymbol{\lambda}^k) - \frac{1}{s}(\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k)\Big], \\ \boldsymbol{z}^{k+1} &= \boldsymbol{z}^k + \alpha(\boldsymbol{x}^{k+1} - \boldsymbol{z}^k), \qquad \boldsymbol{\mu}^{k+1} = \boldsymbol{\mu}^k + \beta(\boldsymbol{\lambda}^{k+1} - \boldsymbol{\mu}^k). \end{aligned} \tag{3.1}$$

PROPOSITION 3.5 (Embedding). *Let $\rho > 0$, and let the sequence $\boldsymbol{w}^k := (\boldsymbol{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{z}^k, \boldsymbol{\mu}^k)$ be generated by $\mathrm{SPLALM}_{\theta,\alpha,\beta}$. Suppose that for some $K \geq 1$, $(\boldsymbol{x}^k,\boldsymbol{\lambda}^k) \in \mathcal{R}_\rho$ and $(\boldsymbol{x}^{k+1},\boldsymbol{\lambda}^k) \in \mathcal{R}_\rho$ for all $0 \leq k < K$. Then (3.1) holds for $0 \leq k < K$, and*

$$R_k = \max\big\{\|\nabla_{\boldsymbol{x}} \Psi_\rho(\boldsymbol{x}^{k+1},\boldsymbol{\lambda}^k)\|,\ \|\nabla_{\boldsymbol{\lambda}} \Psi_\rho(\boldsymbol{x}^{k+1},\boldsymbol{\lambda}^k)\|\big\}, \qquad 0 \leq k < K.$$

Proof. Apply Proposition 3.4 at $(\boldsymbol{x}^k,\boldsymbol{\lambda}^k)$ in the $\boldsymbol{x}$-update of (2.6), and at $(\boldsymbol{x}^{k+1},\boldsymbol{\lambda}^k)$ in the $\boldsymbol{\lambda}$-update. The same point gives $\boldsymbol{y}^{k+1} = \mathbf{0}$ and $\boldsymbol{q}^k = \boldsymbol{\lambda}^k + \rho\mathcal{A}(\boldsymbol{x}^{k+1})$, so the formula for R_k follows from (2.2). $\square$

Recursion (3.1) is the update of DS-GDA [39] for the saddle function Ψ_ρ, with the same order of updates; the parameter correspondence for DS-GDA and its special cases is listed in Table 2.

Remark 3.6. The following points limit the scope of Propositions 3.4 and 3.5.

(i) The projections onto X and Y in the cited methods are inactive when the trajectory stays in the interior of $X \times Y$. For $h = \|\cdot\|_2$, we have $\partial h(\mathbf{0}) = \mathbb{B}(\mathbf{0}, 1)$ and $h^* = \iota_{\mathbb{B}(\mathbf{0},1)}$, so $Y = \mathbb{B}(\mathbf{0}, 1)$ is the natural dual set.

(ii) For a smooth $\tilde{f}$, SPLALM$_{\theta,\alpha,\beta}$ on the instance $\big(\tilde{f} - \frac{\rho}{2}\|\mathcal{A}\|^2, \mathcal{A}, h\big)$ generates (3.1) with $\Psi = \tilde{f} + \langle\cdot, \mathcal{A}\rangle$ whenever its trajectory stays in $\mathcal{R}_\rho$. A hard instance of this form for Smoothed GDA or DS-GDA is therefore a hard instance for SPLALM$_{\theta,\alpha,\beta}$ with any $\rho > 0$, provided that $\tilde{f} - \frac{\rho}{2}\|\mathcal{A}\|^2$ satisfies the bounds in Definition 2.1.

(iii) Outside $\mathcal{R}_\rho$, the two recursions differ. The analysis of SPLALM uses the smoothness of $\mathcal{D}_\rho(\cdot, \boldsymbol{\lambda})$ in (2.3), which requires $\rho > 0$; it does not apply to GDA-type methods.

(iv) The case $s = \infty$ is reproduced by $\beta = 1$ for every finite s; see Section 2.4. PSGDA with $r_y = \Theta(\epsilon)$ has $s = \Theta(\epsilon^{-1})$; whether it satisfies $\epsilon(\rho + s) \le c_1$ depends on the constants in r_y and c_1.

TODO: State the transfer of Theorem 3.1 to Smoothed GDA and DS-GDA as a corollary.

4. Hard Instances and Proof of Theorem 3.1. The proof uses two instances, according to the size of the primal center stepsize α. Set $\varpi := \max\{1, 4\tau, 4\gamma\tau\}$ and $\alpha_\epsilon := 1024\varpi\epsilon^2$.

Small α.. Take $\mathcal{A} \equiv 0$, $h = |\cdot|$, and let f be a one-dimensional Huber function with derivative -2ϵ on a linear interval and initial objective gap Δ. Since $\mathcal{A} \equiv 0$, the multipliers stay at zero, and the center z^k moves by at most $2\alpha\gamma\epsilon$ per iteration, so $x^k \le 2\gamma\epsilon(1 + \alpha k)$. The iterates therefore stay on the plateau, where $R_k = 2\epsilon$, for $\Omega\big(\Delta/(\gamma\alpha\epsilon^2)\big)$ iterations. For $\alpha \le \alpha_\epsilon$, this is $\Omega(\Delta\epsilon^{-4})$.

Figure 1 illustrates the constant-gradient interval and its transition to the quadratic region.

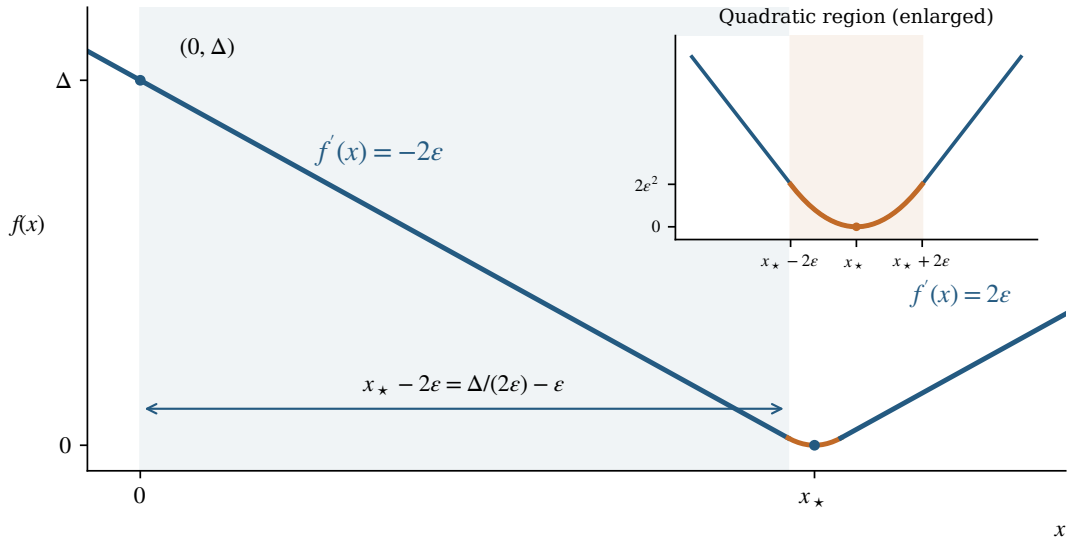


Fig. 1: The one-dimensional Huber instance, illustrated with $\Delta = 1$ and $\epsilon = 0.1$, where $x_\star = \Delta/(2\epsilon) + \epsilon$. The shaded interval $[0, x_\star - 2\epsilon]$ has derivative -2ϵ, and $f(0) = \Delta$. The inset enlarges the quadratic region $[x_\star - 2\epsilon, x_\star + 2\epsilon]$, where $f(x) = \frac{1}{2}(x - x_\star)^2$ and $f(x_\star) = 0$. The two linear tails ensure bounded sublevel sets.

Algorithm 6.1 Smoothed-Perturbed Linearized Augmented Lagrangian Method (SPLALM)

Input: Initial $\boldsymbol{x}^0, \boldsymbol{z}^0 \in \mathbb{R}^n$ and $\boldsymbol{\lambda}^0, \boldsymbol{\mu}^0 \in \mathbb{R}^m$; Parameters: $\rho > 0$ and $\gamma > 0$; sequences: $\{\eta_k\}$, $\{\tau_k\}$, $\{\alpha_k\}$, and $\{\beta_k\}$; and iteration limit K.
for $k = 0, 1, \ldots, K-1$ **do**
  $\boldsymbol{x}^{k+1} = \boldsymbol{x}^k - \eta_k \nabla_{\boldsymbol{x}} \mathcal{P}_{s_k}(\boldsymbol{x}^k, \boldsymbol{\lambda}^k; \boldsymbol{z}^k, \boldsymbol{\mu}^k)$;
  $\boldsymbol{\lambda}^{k+1} = \boldsymbol{\lambda}^k + \tau_k \nabla_{\boldsymbol{\lambda}} \mathcal{P}_{s_k}(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k; \boldsymbol{z}^k, \boldsymbol{\mu}^k)$.
  $\boldsymbol{z}^{k+1} = \boldsymbol{z}^k + \alpha_k(\boldsymbol{x}^{k+1} - \boldsymbol{z}^k)$.
  $\boldsymbol{\mu}^{k+1} = \boldsymbol{\mu}^k + \beta_k(\boldsymbol{\lambda}^{k+1} - \boldsymbol{\mu}^k)$.
end for

Large α.. Take $h = \|\cdot\|_2$, $\mathcal{A}(\boldsymbol{x}) = b(\boldsymbol{x} - \boldsymbol{x}_c)$, and let f be a quadratic with negative curvature $-\kappa$ near $\boldsymbol{x}_c$. While the trajectory stays in $\mathcal{R}_\rho$, Proposition 3.5 shows that the update is the linear time-invariant system $\boldsymbol{w}^{k+1} = \mathbb{T}\boldsymbol{w}^k$. We choose b and κ so that $\mathbb{T}$ has an eigenvalue ζ with $|\zeta| = 1$. Along its eigenvector $\boldsymbol{v}$, the iterates rotate without decay, and the certificate keeps a fixed positive value. The zero initialization is not parallel to $\boldsymbol{v}$. We write it as $\boldsymbol{v} + \boldsymbol{r}$, where $\boldsymbol{r}$ lies in the invariant subspace of the remaining eigenvalues, all strictly inside the unit circle. The shifts $\boldsymbol{x}_c$ and $\boldsymbol{\lambda}_c$ make $\boldsymbol{v} + \boldsymbol{r}$ correspond to the origin. A bound on the transient $\mathbb{T}^k\boldsymbol{r}$ keeps the trajectory in $\mathcal{R}_\rho$ and gives $R_k \geq \frac{7}{6}\epsilon$ for all k. The case $\beta = 0$ uses a separate one-dimensional instance with a Huber outer function h.

TODO: Subsections: plateau instance; rotating instance (parameters, spectral decomposition, transient bound, verification); $\beta = 0$ instance; proof of Theorem 3.1 with constants c_1, c_2.

5. Time-Varying Center Stepsizes. TODO: Lower bound for predetermined center schedules.

6. Upper Bound with Increasing s_k. TODO: Main upper bound within the same family $\mathcal{C}_\theta$: constant parameters with $s = \Theta(1/\epsilon)$.

Throughout this section, we additionally impose the following assumption. It is not part of the problem class $\mathcal{I}(\mathbf{C}, \Delta)$ of Definition 2.1.

Assumption 3 (Level-bounded objective). For every $c \in \mathbb{R}$, the sublevel set $\{\boldsymbol{x} \in \mathbb{R}^n : F(\boldsymbol{x}) \leq c\}$ is empty or compact.

6.1. Algorithm with increasing s_k.

This section uses the updates of Definition 2.6 with iteration-dependent parameters s_k, η_k, τ_k, α_k, and β_k, and with an arbitrary initialization; see Algorithm 6.1. By (2.3)–(2.5), the first two updates read

$$\begin{aligned} \boldsymbol{x}^{k+1} &= \boldsymbol{x}^k - \eta_k\Big[\nabla f(\boldsymbol{x}^k) + J_{\mathcal{A}}(\boldsymbol{x}^k)^\top \boldsymbol{q}(\boldsymbol{x}^k, \boldsymbol{\lambda}^k) + \frac{1}{\gamma}(\boldsymbol{x}^k - \boldsymbol{z}^k)\Big], \\ \boldsymbol{\lambda}^{k+1} &= \boldsymbol{\lambda}^k + \tau_k\Big[\mathcal{A}(\boldsymbol{x}^{k+1}) - \boldsymbol{y}(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k) - \frac{1}{s_k}(\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k)\Big], \end{aligned} \tag{6.1}$$

and the outputs $(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}, \boldsymbol{q}^k)$ are those of Definition 2.6.

We use fixed primal and dual stepsizes together with diminishing center stepsizes. Choose $\rho > 0$ and $\gamma > 0$ satisfying Proposition 6.2. With L_x, μ_γ, and $L_{d,0}$ defined in Section 6.2, set $\eta_k \equiv \eta$ and

$\tau_k \equiv \tau$. An admissible choice is

$$s_k = c_s(k+1)^{1/3}, \qquad \eta = \frac{1}{2L_x}, \qquad \tau = \left(2L_{d,0} + \frac{72L_x G_{\mathcal{A}}^2}{\mu_\gamma^2}\right)^{-1}, \tag{6.2}$$
$$\alpha_k = \frac{\min\{s_0, \gamma\mu_\gamma\tau\}}{36 s_k}, \qquad \beta_k = \frac{\theta_\mu}{k+k_0},$$

where $c_s > 0$, $\theta_\mu \in (0,1]$, and $k_0 \geq 1$ are constants independent of K. Consequently,

$$s_k = \Theta((k+1)^{1/3}), \qquad \alpha_k = \Theta((k+1)^{-1/3}), \qquad \beta_k = \Theta((k+1)^{-1}).$$

THEOREM 6.1. *Suppose that Assumptions 3, 1, and 2 hold. Let the sequence generated by Algorithm 6.1 use (6.2), with ρ and γ as specified above. For $K \geq 2$, choose a tail-best index*

$$\kappa \in \operatorname*{argmin}_{k \in \{\lceil K/2 \rceil, \ldots, K-1\}} \left\{ \|\Delta_{\boldsymbol{x}}^{k+1}\|^2 + \|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2 + \alpha_k \|\boldsymbol{x}^{k+1} - \boldsymbol{z}^k\|^2 \right\}.$$

Then it holds that

$$\max\left\{ \left[\left\| \nabla f(\boldsymbol{x}^{\kappa+1}) + J_{\mathcal{A}}(\boldsymbol{x}^{\kappa+1})^\top \boldsymbol{q}^\kappa \right\|^2 \right], \left[\left\| \mathcal{A}(\boldsymbol{x}^{\kappa+1}) - \boldsymbol{y}^{\kappa+1} \right\|^2 \right] \right\} = \mathcal{O}(K^{-2/3}),$$

with $\boldsymbol{q}^\kappa \in \partial h(\boldsymbol{y}^{\kappa+1})$. Consequently, given $\epsilon > 0$, the iteration complexity of Algorithm 6.1 to obtain an ϵ-KKT point is of order $\mathcal{O}(\epsilon^{-3})$.

6.2. Complexity Analysis. Let us define the constants

$$L_{\mathcal{D},\boldsymbol{x}} := L_f + L_{\mathcal{A}} L_h + \rho G_{\mathcal{A}}^2, \tag{6.3}$$

and

$$\mu_\gamma := \frac{1}{\gamma} - L_{\mathcal{D},\boldsymbol{x}}, \qquad L_x := L_{\mathcal{D},\boldsymbol{x}} + \frac{1}{\gamma}, \qquad L_\lambda(s) := \frac{1}{\rho} + \frac{1}{s},$$

where L_f, $L_{\mathcal{A}}$, L_h, ρ, and $G_{\mathcal{A}}$ defined as before.

The following proposition summarizes the curvature and smoothness properties of the smoothed perturbed potential function $\mathcal{P}_s$. The proof is deferred to Appendix .1.

PROPOSITION 6.2. *Suppose that Assumptions 1 and 2 hold, and let the smoothing parameter $0 < \gamma < \frac{1}{2L_{\mathcal{D},\boldsymbol{x}}}$. Then, for all $s > 0$, $\mathcal{P}_s$ is μ_γ-strongly convex in $\boldsymbol{x}$ and $1/s$-strongly concave in $\boldsymbol{\lambda}$. Moreover, for all $\boldsymbol{x}, \boldsymbol{x}' \in \mathbb{R}^n$, $\boldsymbol{\lambda}, \boldsymbol{\lambda}' \in \mathbb{R}^m$, $\boldsymbol{z} \in \mathbb{R}^n$, and $\boldsymbol{\mu} \in \mathbb{R}^m$, it holds that*

$$\left\| \nabla_{\boldsymbol{x}} \mathcal{P}_s(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{z}, \boldsymbol{\mu}) - \nabla_{\boldsymbol{x}} \mathcal{P}_s(\boldsymbol{x}', \boldsymbol{\lambda}', \boldsymbol{z}, \boldsymbol{\mu}) \right\| \leq L_x \|\boldsymbol{x} - \boldsymbol{x}'\| + G_{\mathcal{A}} \|\boldsymbol{\lambda} - \boldsymbol{\lambda}'\|, \tag{6.4}$$
$$\left\| \nabla_{\boldsymbol{\lambda}} \mathcal{P}_s(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{z}, \boldsymbol{\mu}) - \nabla_{\boldsymbol{\lambda}} \mathcal{P}_s(\boldsymbol{x}', \boldsymbol{\lambda}', \boldsymbol{z}, \boldsymbol{\mu}) \right\| \leq G_{\mathcal{A}} \|\boldsymbol{x} - \boldsymbol{x}'\| + L_\lambda(s) \|\boldsymbol{\lambda} - \boldsymbol{\lambda}'\|. \tag{6.5}$$

In particular, $\mathcal{P}_s$ is L_x-smooth in $\boldsymbol{x}$ and $L_\lambda(s)$-smooth in $\boldsymbol{\lambda}$.

6.2.1. Value Functions and Lyapunov Function. Fix $s > 0$ and $(\boldsymbol{z}, \boldsymbol{\mu})$. By Proposition 6.2, $\mathcal{P}_s(\cdot, \cdot; \boldsymbol{z}, \boldsymbol{\mu})$ is strongly convex in $\boldsymbol{x}$ and strongly concave in $\boldsymbol{\lambda}$. Hence the following value functions are well defined, and the minimizer and maximizer below are unique:

$$d_s(\boldsymbol{\lambda}; \boldsymbol{z}, \boldsymbol{\mu}) := \min_{\boldsymbol{x} \in \mathbb{R}^n} \mathcal{P}_s(\boldsymbol{x}, \boldsymbol{\lambda}; \boldsymbol{z}, \boldsymbol{\mu}), \qquad \boldsymbol{x}_s(\boldsymbol{\lambda}; \boldsymbol{z}, \boldsymbol{\mu}) := \operatorname*{argmin}_{\boldsymbol{x} \in \mathbb{R}^n} \mathcal{P}_s(\boldsymbol{x}, \boldsymbol{\lambda}; \boldsymbol{z}, \boldsymbol{\mu}), \tag{6.6}$$

$$p_s(\boldsymbol{z},\boldsymbol{\mu}) := \max_{\boldsymbol{\lambda}\in\mathbb{R}^m} d_s(\boldsymbol{\lambda};\boldsymbol{z},\boldsymbol{\mu}), \qquad \boldsymbol{\lambda}_s^\star(\boldsymbol{z},\boldsymbol{\mu}) := \operatorname*{argmax}_{\boldsymbol{\lambda}\in\mathbb{R}^m} d_s(\boldsymbol{\lambda};\boldsymbol{z},\boldsymbol{\mu}). \tag{6.7}$$

Strong convexity–strong concavity also yields strong duality,

$$p_s(\boldsymbol{z},\boldsymbol{\mu}) = \min_{\boldsymbol{x}\in\mathbb{R}^n}\max_{\boldsymbol{\lambda}\in\mathbb{R}^m} \mathcal{P}_s(\boldsymbol{x},\boldsymbol{\lambda};\boldsymbol{z},\boldsymbol{\mu}), \tag{6.8}$$

and the unique saddle point is $(\boldsymbol{x}_s^\star(\boldsymbol{z},\boldsymbol{\mu}),\boldsymbol{\lambda}_s^\star(\boldsymbol{z},\boldsymbol{\mu}))$, where

$$\boldsymbol{x}_s^\star(\boldsymbol{z},\boldsymbol{\mu}) := \boldsymbol{x}_s\big(\boldsymbol{\lambda}_s^\star(\boldsymbol{z},\boldsymbol{\mu});\boldsymbol{z},\boldsymbol{\mu}\big).$$

Following the three-value Lyapunov construction for smoothed nonconvex–concave minimax methods [37], and its perturbed-smoothed extension [14], we adapt the construction to the reduced augmented-Lagrangian setting and define

$$\begin{aligned}\Psi_s(\boldsymbol{x},\boldsymbol{\lambda};\boldsymbol{z},\boldsymbol{\mu}) &:= \mathcal{P}_s(\boldsymbol{x},\boldsymbol{\lambda};\boldsymbol{z},\boldsymbol{\mu}) - 2d_s(\boldsymbol{\lambda};\boldsymbol{z},\boldsymbol{\mu}) + 2p_s(\boldsymbol{z},\boldsymbol{\mu})\\ &= \underbrace{\mathcal{P}_s(\boldsymbol{x},\boldsymbol{\lambda};\boldsymbol{z},\boldsymbol{\mu}) - d_s(\boldsymbol{\lambda};\boldsymbol{z},\boldsymbol{\mu})}_{\text{primal gap at fixed }\boldsymbol{\lambda}} + \underbrace{p_s(\boldsymbol{z},\boldsymbol{\mu}) - d_s(\boldsymbol{\lambda};\boldsymbol{z},\boldsymbol{\mu})}_{\text{dual gap}} + p_s(\boldsymbol{z},\boldsymbol{\mu}).\end{aligned} \tag{6.9}$$

Both gaps are nonnegative, so $\Psi_s \ge p_s$.

Iteration-k shorthand.. At iteration k we evaluate the above objects at $(s,\boldsymbol{\mu}) = (s_k,\boldsymbol{\mu}^k)$, replace the subscript s by k, and suppress $\boldsymbol{\mu}^k$. For example, $\mathcal{P}_k(\boldsymbol{x},\boldsymbol{\lambda};\boldsymbol{z}) := \mathcal{P}_{s_k}(\boldsymbol{x},\boldsymbol{\lambda};\boldsymbol{z},\boldsymbol{\mu}^k)$ and $d_k(\boldsymbol{\lambda};\boldsymbol{z}) := d_{s_k}(\boldsymbol{\lambda};\boldsymbol{z},\boldsymbol{\mu}^k)$; the quantities p_k, $\boldsymbol{x}_k$, $\boldsymbol{\lambda}_k^\star$, $\boldsymbol{x}_k^\star$, and Ψ_k are defined analogously. We also write $\Delta_{\boldsymbol{x}}^{k+1} := \boldsymbol{x}^{k+1} - \boldsymbol{x}^k$, and define $\Delta_{\boldsymbol{\lambda}}^{k+1}$, $\Delta_{\boldsymbol{z}}^{k+1}$, and $\Delta_{\boldsymbol{\mu}}^{k+1}$ in the same way.

6.2.2. Basic One-Step Estimate. Throughout this subsection, we assume that

$$s_{k+1} \ge s_k > 0, \qquad 0 < \tau_k \le s_k, \qquad 0 < \beta_k \le 1, \qquad 0 \le \alpha_k \le 2. \tag{6.10}$$

The following lemma gives a basic one-step descent estimate of the potential function $\mathcal{P}$ from iteration k to $k+1$.

LEMMA 6.3. *Suppose Assumptions 1 and 2 hold. Let the iterates be generated by Algorithm 6.1 with the parameters satisfying (6.10). Then, for every $k \ge 0$, the following holds:*

$$\begin{aligned}&\mathcal{P}_{k+1}(\boldsymbol{x}^{k+1},\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^{k+1}) - \mathcal{P}_k(\boldsymbol{x}^k,\boldsymbol{\lambda}^k;\boldsymbol{z}^k)\\ &\le -\left(\frac{1}{\eta_k} - \frac{L_x}{2}\right)\|\Delta_{\boldsymbol{x}}^{k+1}\|^2 + \left(\frac{1}{\tau_k} - \frac{1}{2s_k}\right)\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2\\ &\quad - \frac{\alpha_k(2-\alpha_k)}{2\gamma}\|\boldsymbol{x}^{k+1} - \boldsymbol{z}^k\|^2 + \frac{\beta_k(2-\beta_k)}{2s_k}\|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\mu}^k\|^2 + \frac{1}{2}\left(\frac{1}{s_k} - \frac{1}{s_{k+1}}\right)\|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\mu}^{k+1}\|^2.\end{aligned} \tag{6.11}$$

As shown in Lemma 6.3, the positive ascent term generated by the $\boldsymbol{\lambda}$-update prevents the potential $\mathcal{P}_k$ from yielding a direct descent estimate. We therefore relate this term to the one-step increase of the dual value function, as developed in Lemma 6.5. This relation allows us to construct a Lyapunov function with a descent property. Before proceeding, we establish several regularity properties of the value functions d_k and p_k, with their gradient formulas following from Danskin's theorem.

LEMMA 6.4. *Under the conditions of Proposition 6.2, for all $k \ge 0$, the following properties hold:*
(i) For every $\boldsymbol{z} \in \mathbb{R}^n$, the minimizer $\boldsymbol{x}_k(\boldsymbol{\lambda};\boldsymbol{z})$ is unique and $\frac{G_A}{\mu_\gamma}$-Lipschitz continuous in $\boldsymbol{\lambda}$.

(ii) for every $\boldsymbol{z} \in \mathbb{R}^n$, $d_k(\cdot;\boldsymbol{z})$ is $1/s_k$-strongly concave and $L_{d,k}$-smooth, with gradient

$$\nabla_{\boldsymbol{\lambda}} d_k(\boldsymbol{\lambda};\boldsymbol{z}) = \nabla_{\boldsymbol{\lambda}} \mathcal{P}_k\big(\boldsymbol{x}_k(\boldsymbol{\lambda};\boldsymbol{z}),\boldsymbol{\lambda};\boldsymbol{z}\big),$$

where $L_{d,k} := \frac{1}{\rho} + \frac{1}{s_k} + \frac{G_{\mathcal{A}}^2}{\mu_\gamma}$.

(iii) The function $p_k(z)$ is continuously differentiable, with gradient

$$\nabla p_k(\boldsymbol{z}) = \frac{1}{\gamma}\big(\boldsymbol{z} - \boldsymbol{x}_k^\star(\boldsymbol{z})\big),$$

and has a $1/\gamma$-Lipschitz continuous gradient.

The following lemma controls the dual ascent term.

LEMMA 6.5. *Suppose that the conditions of Lemma 6.4 hold, and let the iterates be generated by Algorithm 6.1 with parameters satisfying (6.10). For every $k \geq 0$, assume that*

$$\frac{1}{\tau_k} + \frac{1}{2s_k} \geq 2L_{d,k}. \tag{6.12}$$

Then the following holds

$$\begin{aligned}&\left(\frac{1}{\tau_k} - \frac{1}{2s_k}\right)\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2 \\ &\leq 2\big[d_k(\boldsymbol{\lambda}^{k+1};\mathbf{z}^k) - d_k(\boldsymbol{\lambda}^k;\mathbf{z}^k)\big] - \frac{L_{d,k}}{2}\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2 + \frac{4G_{\mathcal{A}}^2}{L_{d,k}\mu_\gamma}\left[\mathcal{P}_k(\mathbf{x}^{k+1},\boldsymbol{\lambda}^k;\mathbf{z}^k) - d_k(\boldsymbol{\lambda}^k;\mathbf{z}^k)\right].\end{aligned}$$

Moreover, it holds that

$$\begin{aligned}&\mathcal{P}_{k+1}(\boldsymbol{x}^{k+1},\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^{k+1}) - \mathcal{P}_k(\boldsymbol{x}^k,\boldsymbol{\lambda}^k;\boldsymbol{z}^k) \\ &\leq -\left[\frac{1}{\eta_k} - \frac{L_x}{2} - \frac{2G_{\mathcal{A}}^2(1/\eta_k + L_x)^2}{L_{d,k}\mu_\gamma^2}\right]\|\Delta_{\boldsymbol{x}}^{k+1}\|^2 - \frac{L_{d,k}}{2}\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2 \\ &\quad + 2\big[d_k(\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^k) - d_k(\boldsymbol{\lambda}^k;\boldsymbol{z}^k)\big] - \frac{\alpha_k(2-\alpha_k)}{2\gamma}\|\boldsymbol{x}^{k+1} - \boldsymbol{z}^k\|^2 \\ &\quad + \frac{\beta_k(2-\beta_k)}{2s_k}\|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\mu}^k\|^2 + \frac{1}{2}\left(\frac{1}{s_k} - \frac{1}{s_{k+1}}\right)\|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\mu}^{k+1}\|^2.\end{aligned} \tag{6.13}$$

The potential $\mathcal{P}_k$ increases along the dual update, so its variation alone does not capture progress in both variables. We therefore seek a merit function that accounts for both primal suboptimality and progress in dual maximization. To account for these two effects, consider the partial potential

$$\mathcal{P}_k - cd_k = (\mathcal{P}_k - d_k) - (c-1)d_k, \qquad c > 1. \tag{6.14}$$

The first term measures the primal optimality gap, while the second term decreases as the dual value increases. In the following analysis, we set $c = 2$, consistent with the coefficient of the dual-value increment in (6.13).

LEMMA 6.6. *Under the conditions of Lemma 6.5, for every $k \geq 0$, the following holds*

$$\begin{aligned}
&\left[\mathcal{P}_{k+1}(\boldsymbol{x}^{k+1},\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^{k+1}) - 2d_k(\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^{k+1})\right] - \left[\mathcal{P}_k(\boldsymbol{x}^k,\boldsymbol{\lambda}^k;\boldsymbol{z}^k) - 2d_k(\boldsymbol{\lambda}^k;\boldsymbol{z}^k)\right] \\
\leq& -\left[\frac{1}{\eta_k} - \frac{L_x}{2} - \frac{2G_{\mathcal{A}}^2(1/\eta_k + L_x)^2}{L_{d,k}\mu_\gamma^2}\right]\|\Delta_{\boldsymbol{x}}^{k+1}\|^2 - \frac{L_{d,k}}{2}\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2 \\
&+ \frac{\alpha_k(2-\alpha_k)}{2\gamma}\|\boldsymbol{x}^{k+1} - \boldsymbol{z}^k\|^2 - \frac{2\alpha_k}{\gamma}\left\langle \boldsymbol{x}^{k+1} - \boldsymbol{x}_k(\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^{k+1}), \boldsymbol{x}^{k+1} - \boldsymbol{z}^k\right\rangle \\
&+ \frac{\beta_k(2-\beta_k)}{2s_k}\|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\mu}^k\|^2 + \frac{1}{2}\left(\frac{1}{s_k} - \frac{1}{s_{k+1}}\right)\|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\mu}^{k+1}\|^2.
\end{aligned} \tag{6.15}$$

Equation (6.15) gives the one-step estimate for the partial potential in (6.14) with $c = 2$. The $\boldsymbol{x}$- and $\boldsymbol{\lambda}$-updates contribute the first two terms on the right-hand side. The $\boldsymbol{z}$-update, however, produces a positive quadratic term and an indefinite inner-product term. Hence, the partial potential does not yet provide a sufficient descent estimate for the full iteration.

At an exact saddle point of $\min_{\boldsymbol{x}} \max_{\boldsymbol{\lambda}} \mathcal{P}_k(\boldsymbol{x},\boldsymbol{\lambda};\boldsymbol{z})$, strong duality gives

$$\mathcal{P}_k\big(\boldsymbol{x}_k^\star(\boldsymbol{z}),\boldsymbol{\lambda}_k^\star(\boldsymbol{z});\boldsymbol{z}\big) = d_k\big(\boldsymbol{\lambda}_k^\star(\boldsymbol{z});\boldsymbol{z}\big) = p_k(\boldsymbol{z}),$$

and therefore

$$\mathcal{P}_k\big(\boldsymbol{x}_k^\star(\boldsymbol{z}),\boldsymbol{\lambda}_k^\star(\boldsymbol{z});\boldsymbol{z}\big) - 2d_k\big(\boldsymbol{\lambda}_k^\star(\boldsymbol{z});\boldsymbol{z}\big) = -p_k(\boldsymbol{z}).$$

Consequently, any $\boldsymbol{z}$-update that decreases the saddle value p_k increases the partial potential $\mathcal{P}_k - 2d_k$. This motivates us to examine whether an additional value-function term can restore a descent estimate.

We therefore add the value-function increment to the partial potential in (6.14): $2[p_k(\boldsymbol{z}^{k+1}) - p_k(\boldsymbol{z}^k)]$ and combine the two estimates directly. The resulting one-step bound is stated next.

LEMMA 6.7. *Under the conditions of Lemma 6.4 and $0 \leq \alpha_k \leq 2$, the $\boldsymbol{z}$-update satisfies*

$$p_k(\boldsymbol{z}^{k+1}) - p_k(\boldsymbol{z}^k) \leq -\frac{\alpha_k(2-\alpha_k)}{2\gamma}\|\boldsymbol{x}^{k+1} - \boldsymbol{z}^k\|^2 + \frac{\alpha_k}{\gamma}\left\langle \boldsymbol{x}^{k+1} - \boldsymbol{x}_k^\star(\boldsymbol{z}^k), \boldsymbol{x}^{k+1} - \boldsymbol{z}^k\right\rangle. \tag{6.16}$$

LEMMA 6.8. *Under the conditions of Lemma 6.6, for every $k \geq 0$,*

$$\begin{aligned}
&\left[\mathcal{P}_{k+1}(\boldsymbol{x}^{k+1},\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^{k+1}) - 2d_k(\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^{k+1}) + 2p_k(\boldsymbol{z}^{k+1})\right] \\
&\quad - \left[\mathcal{P}_k(\boldsymbol{x}^k,\boldsymbol{\lambda}^k;\boldsymbol{z}^k) - 2d_k(\boldsymbol{\lambda}^k;\boldsymbol{z}^k) + 2p_k(\boldsymbol{z}^k)\right] \\
\leq& -\left[\frac{1}{\eta_k} - \frac{L_x}{2} - \frac{2G_{\mathcal{A}}^2(1/\eta_k + L_x)^2}{L_{d,k}\mu_\gamma^2}\right]\|\Delta_{\boldsymbol{x}}^{k+1}\|^2 - \frac{L_{d,k}}{2}\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2 \\
&- \frac{\alpha_k(2-\alpha_k)}{2\gamma}\|\boldsymbol{x}^{k+1} - \boldsymbol{z}^k\|^2 + \frac{2\alpha_k}{\gamma}\left\langle \boldsymbol{x}_k(\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^{k+1}) - \boldsymbol{x}_k^\star(\boldsymbol{z}^k), \boldsymbol{x}^{k+1} - \boldsymbol{z}^k\right\rangle \\
&+ \frac{\beta_k(2-\beta_k)}{2s_k}\|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\mu}^k\|^2 + \frac{1}{2}\left(\frac{1}{s_k} - \frac{1}{s_{k+1}}\right)\|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\mu}^{k+1}\|^2.
\end{aligned} \tag{6.17}$$

Two gaps remain before Φ_k admits a genuine descent inequality: the augmented estimate still carries an indefinite cross term and positive multiplier/center-increment terms, and its left-hand side compares d_k, p_k at both endpoints rather than the true $k \to k+1$ potential difference, which requires d_{k+1}, p_{k+1}.

For that, we establish boundedness of the multipliers directly, rather than assuming compactness a priori: both multipliers are controlled by the AL structure and the inner residual, not by projection onto a compact set.

LEMMA 6.9. *Suppose Assumptions 1 and 2 hold. Let the iterates be generated by Algorithm 6.1 with (6.10) and*

$$\tau_k \left(\frac{1}{\rho} + \frac{1}{s_k} \right) \leq 1. \tag{6.18}$$

Define the constant independent of k

$$M := \max\{\|\boldsymbol{\lambda}^0\|, \|\boldsymbol{\mu}^0\|, L_h\}.$$

Then, for every $k \geq 0$, the following holds

$$\|\boldsymbol{\lambda}^k\| \leq M, \qquad \|\boldsymbol{\mu}^k\| \leq M.$$

Moreover, for every $k \geq 0$ and every $\boldsymbol{z} \in \mathbb{R}^n$, it holds that

$$\|\boldsymbol{\lambda}_k^\star(\boldsymbol{z})\| \leq M.$$

LEMMA 6.10. *Under the conditions of Proposition 6.2, for every $k \geq 0$, $\boldsymbol{\lambda} \in \mathbb{R}^m$, and $\boldsymbol{z} \in \mathbb{R}^n$, it holds that*

$$\|\boldsymbol{x}_k(\boldsymbol{\lambda}; \boldsymbol{z}) - \boldsymbol{x}_k^\star(\boldsymbol{z})\|^2 \leq \frac{2}{\mu_\gamma} \left[p_k(\boldsymbol{z}) - d_k(\boldsymbol{\lambda}; \boldsymbol{z}) \right]. \tag{6.19}$$

For any $\boldsymbol{z}, \boldsymbol{z}' \in \mathbb{R}^n$,

$$\max\left\{ \|\boldsymbol{x}_k(\boldsymbol{\lambda}; \boldsymbol{z}') - \boldsymbol{x}_k(\boldsymbol{\lambda}; \boldsymbol{z})\|, \|\boldsymbol{x}_k^\star(\boldsymbol{z}') - \boldsymbol{x}_k^\star(\boldsymbol{z})\| \right\} \leq \frac{1}{\gamma \mu_\gamma} \|\boldsymbol{z}' - \boldsymbol{z}\|. \tag{6.20}$$

We now establish an approximate descent inequality for Algorithm 6.1 with respect to the Lyapunov function defined in (6.9). We assume the parameters are chosen as

$$\begin{aligned} &0 < \gamma < \frac{1}{2L_{\mathcal{D},\boldsymbol{x}}}, \qquad 0 < s_k \leq s_{k+1}, \qquad 0 < \beta_k \leq 1, \qquad 0 < \eta_k \leq \frac{1}{2L_x}, \\ &0 < \tau_k \leq \left(\max\left\{ 2L_{d,k}, \frac{36 G_{\mathcal{A}}^2}{\eta_k \mu_\gamma^2} \right\} \right)^{-1}, \qquad 0 < \alpha_k \leq \frac{1}{36} \min\left\{ 1, \frac{\gamma \mu_\gamma \tau_k}{s_k} \right\}, \end{aligned} \tag{6.21}$$

where $L_{d,k} := \rho^{-1} + s_k^{-1} + G_{\mathcal{A}}^2/\mu_\gamma$. Since $L_{d,k} \geq \rho^{-1} + s_k^{-1}$, (6.21) implies the conditions of Proposition 6.2, Lemma 6.9, (6.10), (6.12), and (6.18).

LEMMA 6.11. *Suppose that Assumptions 3, 1, and 2 hold, and let $\{(\boldsymbol{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{z}^k, \boldsymbol{\mu}^k)\}$ be generated by Algorithm 6.1 with parameters satisfying (6.21). Then for every $k \geq 0$:*

(a) for all $\boldsymbol{x} \in \mathbb{R}^n$, $\boldsymbol{\lambda} \in \mathbb{R}^m$, and $\boldsymbol{z} \in \mathbb{R}^n$,

$$\Psi_k(\boldsymbol{x}, \boldsymbol{\lambda}; \boldsymbol{z}) \geq p_k(\boldsymbol{z}) \geq \inf_{\boldsymbol{x} \in \mathbb{R}^n} F(\boldsymbol{x}) - \frac{(L_h + M)^2}{2 s_0}; \tag{6.22}$$

(b) the iterates satisfy

$$\begin{aligned}\Psi_{k+1}(\boldsymbol{x}^{k+1},\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^{k+1}) - \Psi_k(\boldsymbol{x}^k,\boldsymbol{\lambda}^k;\boldsymbol{z}^k)\\ \le -\frac{1}{4\eta_k}\|\Delta_{\boldsymbol{x}}^{k+1}\|^2 - \frac{1}{8\tau_k}\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2 - \frac{\alpha_k}{2\gamma}\|\boldsymbol{x}^{k+1}-\boldsymbol{z}^k\|^2 + \delta_k,\end{aligned} \tag{6.23}$$

where $\delta_k := 10M^2\big(s_k^{-1} - s_{k+1}^{-1}\big) + 20M^2\beta_k/s_k$.

The proof is given in Appendix .1.

.1. Proofs of the Auxiliary Results.

Proof of the curvature part of Proposition 6.2. For fixed $\boldsymbol{\lambda}$, the reduced payoff satisfies

$$\nabla_{\boldsymbol{x}}\mathcal{D}_\rho(\boldsymbol{x},\boldsymbol{\lambda}) = \nabla f(\boldsymbol{x}) + J_{\mathcal{A}}(\boldsymbol{x})^\top \boldsymbol{q}(\boldsymbol{x},\boldsymbol{\lambda}),$$

where

$$\boldsymbol{q}(\boldsymbol{x},\boldsymbol{\lambda}) = \operatorname{prox}_{\rho h^*}\big(\boldsymbol{\lambda} + \rho\mathcal{A}(\boldsymbol{x})\big).$$

The nonexpansiveness of the proximal mapping, the bounded Jacobian, and the Lipschitz Jacobian imply

$$\|\boldsymbol{q}(\boldsymbol{x},\boldsymbol{\lambda}) - \boldsymbol{q}(\boldsymbol{x}',\boldsymbol{\lambda})\| \le \rho G_{\mathcal{A}}\|\boldsymbol{x}-\boldsymbol{x}'\|.$$

Assumption 2 gives $\|\boldsymbol{q}(\boldsymbol{x},\boldsymbol{\lambda})\| \le L_h$. Therefore

$$\|\nabla_{\boldsymbol{x}}\mathcal{D}_\rho(\boldsymbol{x},\boldsymbol{\lambda}) - \nabla_{\boldsymbol{x}}\mathcal{D}_\rho(\boldsymbol{x}',\boldsymbol{\lambda})\| \le L_{\mathcal{D},\boldsymbol{x}}\|\boldsymbol{x}-\boldsymbol{x}'\|.$$

Thus $\mathcal{D}_\rho(\cdot,\boldsymbol{\lambda})$ is $L_{\mathcal{D},\boldsymbol{x}}$-weakly convex. Adding $\frac{1}{2\gamma}\|\boldsymbol{x}-\boldsymbol{z}\|^2$ gives μ_γ-strong convexity in $\boldsymbol{x}$.

For fixed $(\boldsymbol{x},\boldsymbol{z},\boldsymbol{\mu})$, $\mathcal{D}_\rho(\boldsymbol{x},\cdot)$ is concave in $\boldsymbol{\lambda}$ because it is the pointwise infimum of affine functions of $\boldsymbol{\lambda}$. The term $-\frac{1}{2s}\|\boldsymbol{\lambda}-\boldsymbol{\mu}\|^2$ is $1/s$-strongly concave. Hence $\mathcal{P}_s$ is $1/s$-strongly concave in $\boldsymbol{\lambda}$. ∎

Proof of the smoothness part of Proposition 6.2. The $\boldsymbol{x}$-block bound follows from the Lipschitz estimate for $\nabla_{\boldsymbol{x}}\mathcal{D}_\rho$ in the proof of Proposition 6.2 and the quadratic term $\frac{1}{2\gamma}\|\boldsymbol{x}-\boldsymbol{z}\|^2$.

For the $\boldsymbol{\lambda}$-block, the identity

$$\nabla_{\boldsymbol{\lambda}}\mathcal{P}_s(\boldsymbol{x},\boldsymbol{\lambda};\boldsymbol{z},\boldsymbol{\mu}) = \frac{1}{\rho}\big(\boldsymbol{q}(\boldsymbol{x},\boldsymbol{\lambda}) - \boldsymbol{\lambda}\big) - \frac{1}{s}(\boldsymbol{\lambda}-\boldsymbol{\mu})$$

and the nonexpansiveness of $\operatorname{prox}_{\rho h^*}$ give $L_\lambda(s) = 1/\rho + 1/s$. The same estimate, together with

$$\|\mathcal{A}(\boldsymbol{x}) - \mathcal{A}(\boldsymbol{x}')\| \le G_{\mathcal{A}}\|\boldsymbol{x}-\boldsymbol{x}'\|,$$

gives both cross Lipschitz bounds with constant $G_{\mathcal{A}}$. ∎

Proof of Lemma 6.5. Set

$$g^k := \nabla_{\boldsymbol{\lambda}}\mathcal{P}_k(\boldsymbol{x}^{k+1},\boldsymbol{\lambda}^k,\boldsymbol{z}^k), \qquad \Delta_{\boldsymbol{\lambda}}^{k+1} = \tau_k g^k,$$

where the second identity follows from the dual update. By the $L_{d,k}$-smoothness of $d_k(\cdot;\boldsymbol{z}^k)$,

$$\begin{aligned}d_k(\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^k) - d_k(\boldsymbol{\lambda}^k;\boldsymbol{z}^k)\\ \ge \big\langle \nabla_{\boldsymbol{\lambda}} d_k(\boldsymbol{\lambda}^k;\boldsymbol{z}^k), \Delta_{\boldsymbol{\lambda}}^{k+1}\big\rangle - \frac{L_{d,k}}{2}\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2.\end{aligned}$$

Danskin's theorem and Lemma 6.4 give

$$\nabla_{\boldsymbol{\lambda}} d_k(\boldsymbol{\lambda}^k; \boldsymbol{z}^k) = \nabla_{\boldsymbol{\lambda}} \mathcal{P}_k\big(\boldsymbol{x}_k(\boldsymbol{\lambda}^k; \boldsymbol{z}^k), \boldsymbol{\lambda}^k, \boldsymbol{z}^k\big).$$

The cross-Lipschitz estimate in Proposition 6.2 therefore yields

$$\begin{aligned}
&\big\langle \nabla_{\boldsymbol{\lambda}} d_k(\boldsymbol{\lambda}^k; \boldsymbol{z}^k), \Delta_{\boldsymbol{\lambda}}^{k+1} \big\rangle \\
&\quad = \tau_k \big\langle \nabla_{\boldsymbol{\lambda}} \mathcal{P}_k\big(\boldsymbol{x}_k(\boldsymbol{\lambda}^k; \boldsymbol{z}^k), \boldsymbol{\lambda}^k, \boldsymbol{z}^k\big), g^k \big\rangle \\
&\quad \geq \tau_k \|g^k\|^2 - \tau_k G_{\mathcal{A}} \|\boldsymbol{x}^{k+1} - \boldsymbol{x}_k(\boldsymbol{\lambda}^k; \boldsymbol{z}^k)\| \|g^k\|.
\end{aligned}$$

Using

$$\tau_k \|g^k\|^2 = \frac{1}{\tau_k} \|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2, \qquad \tau_k \|g^k\| = \|\Delta_{\boldsymbol{\lambda}}^{k+1}\|,$$

we obtain

$$\begin{aligned}
&d_k(\boldsymbol{\lambda}^{k+1}; \boldsymbol{z}^k) - d_k(\boldsymbol{\lambda}^k; \boldsymbol{z}^k) \\
&\quad \geq \left(\frac{1}{\tau_k} - \frac{L_{d,k}}{2} \right) \|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2 - G_{\mathcal{A}} \|\boldsymbol{x}^{k+1} - \boldsymbol{x}_k(\boldsymbol{\lambda}^k; \boldsymbol{z}^k)\| \|\Delta_{\boldsymbol{\lambda}}^{k+1}\|.
\end{aligned} \tag{.24}$$

Let

$$b_k := \frac{1}{\tau_k} + \frac{1}{2s_k} - L_{d,k}.$$

Multiplying (.24) by 2 and rearranging gives

$$\begin{aligned}
&\left(\frac{1}{\tau_k} - \frac{1}{2s_k} \right) \|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2 \\
&\quad \leq 2\big[d_k(\boldsymbol{\lambda}^{k+1}; \boldsymbol{z}^k) - d_k(\boldsymbol{\lambda}^k; \boldsymbol{z}^k) \big] - b_k \|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2 \\
&\qquad + 2G_{\mathcal{A}} \|\boldsymbol{x}^{k+1} - \boldsymbol{x}_k(\boldsymbol{\lambda}^k; \boldsymbol{z}^k)\| \|\Delta_{\boldsymbol{\lambda}}^{k+1}\|.
\end{aligned}$$

The assumed parameter condition implies $b_k \geq L_{d,k} > 0$. Young's inequality with parameter $L_{d,k}$ gives

$$2G_{\mathcal{A}} \|\boldsymbol{x}^{k+1} - \boldsymbol{x}_k(\boldsymbol{\lambda}^k; \boldsymbol{z}^k)\| \|\Delta_{\boldsymbol{\lambda}}^{k+1}\| \leq \frac{L_{d,k}}{2} \|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2 + \frac{2G_{\mathcal{A}}^2}{L_{d,k}} \|\boldsymbol{x}^{k+1} - \boldsymbol{x}_k(\boldsymbol{\lambda}^k; \boldsymbol{z}^k)\|^2.$$

Since $b_k \geq L_{d,k}$, we have

$$-b_k \|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2 + \frac{L_{d,k}}{2} \|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2 \leq -\frac{L_{d,k}}{2} \|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2.$$

Finally, the μ_γ-strong convexity of $\mathcal{P}_k(\cdot, \boldsymbol{\lambda}^k; \boldsymbol{z}^k)$ gives

$$\|\boldsymbol{x}^{k+1} - \boldsymbol{x}_k(\boldsymbol{\lambda}^k; \boldsymbol{z}^k)\|^2 \leq \frac{2}{\mu_\gamma} \left[\mathcal{P}_k(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k; \boldsymbol{z}^k) - d_k(\boldsymbol{\lambda}^k; \boldsymbol{z}^k) \right].$$

Substituting this estimate proves the first inequality. ∎

Proof of the second estimate in Lemma 6.5. Fix $k \geq 0$. By Proposition 6.2, $\mathcal{P}_k(\cdot, \boldsymbol{\lambda}^k; \boldsymbol{z}^k)$ is μ_γ-strongly convex. Applying the strong-convexity inequality at $\boldsymbol{x}^{k+1}$ to the minimizer $\boldsymbol{x}_k(\boldsymbol{\lambda}^k; \boldsymbol{z}^k)$ gives

$$\begin{aligned}
&\mathcal{P}_k(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k; \boldsymbol{z}^k) - d_k(\boldsymbol{\lambda}^k; \boldsymbol{z}^k) \\
&\quad \leq \left\langle \nabla_{\boldsymbol{x}} \mathcal{P}_k(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k; \boldsymbol{z}^k), \boldsymbol{x}^{k+1} - \boldsymbol{x}_k(\boldsymbol{\lambda}^k; \boldsymbol{z}^k) \right\rangle \\
&\qquad - \frac{\mu_\gamma}{2} \|\boldsymbol{x}^{k+1} - \boldsymbol{x}_k(\boldsymbol{\lambda}^k; \boldsymbol{z}^k)\|^2 \\
&\quad \leq \frac{1}{2\mu_\gamma} \|\nabla_{\boldsymbol{x}} \mathcal{P}_k(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k; \boldsymbol{z}^k)\|^2,
\end{aligned}$$

where the last inequality follows from Young's inequality. The $\boldsymbol{x}$-update and the L_x-Lipschitz continuity of $\nabla_{\boldsymbol{x}} \mathcal{P}_k(\cdot, \boldsymbol{\lambda}^k; \boldsymbol{z}^k)$ imply

$$\begin{aligned}
&\|\nabla_{\boldsymbol{x}} \mathcal{P}_k(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k; \boldsymbol{z}^k)\| \\
&\quad \leq \|\nabla_{\boldsymbol{x}} \mathcal{P}_k(\boldsymbol{x}^k, \boldsymbol{\lambda}^k; \boldsymbol{z}^k)\| + L_x \|\Delta_{\boldsymbol{x}}^{k+1}\| \\
&\quad = \left(\frac{1}{\eta_k} + L_x \right) \|\Delta_{\boldsymbol{x}}^{k+1}\|.
\end{aligned}$$

Combining these estimates yields

$$\mathcal{P}_k(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k; \boldsymbol{z}^k) - d_k(\boldsymbol{\lambda}^k; \boldsymbol{z}^k) \leq \frac{(1/\eta_k + L_x)^2}{2\mu_\gamma} \|\Delta_{\boldsymbol{x}}^{k+1}\|^2. \tag{.25}$$

Substituting (.25) into the first inequality of Lemma 6.5 gives a bound on the dual ascent term, where the coefficient of $\|\Delta_{\boldsymbol{x}}^{k+1}\|^2$ becomes

$$\frac{4G_{\mathcal{A}}^2}{L_{d,k}\mu_\gamma} \frac{(1/\eta_k + L_x)^2}{2\mu_\gamma} = \frac{2G_{\mathcal{A}}^2 (1/\eta_k + L_x)^2}{L_{d,k}\mu_\gamma^2}.$$

Substitute the resulting bound on $(1/\tau_k - 1/(2s_k))\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2$ into (6.11) and combine the coefficients of $\|\Delta_{\boldsymbol{x}}^{k+1}\|^2$. The $\boldsymbol{z}$-step term and both parameter-drift terms remain unchanged. This proves (6.13). ∎

Proof of Lemma 6.6. Subtract $2[d_k(\boldsymbol{\lambda}^{k+1}; \boldsymbol{z}^k) - d_k(\boldsymbol{\lambda}^k; \boldsymbol{z}^k)]$ from both sides of (6.13). To replace $d_k(\boldsymbol{\lambda}^{k+1}; \boldsymbol{z}^k)$ on the left by $d_k(\boldsymbol{\lambda}^{k+1}; \boldsymbol{z}^{k+1})$, add $-2[d_k(\boldsymbol{\lambda}^{k+1}; \boldsymbol{z}^{k+1}) - d_k(\boldsymbol{\lambda}^{k+1}; \boldsymbol{z}^k)]$ to both sides. By the minimum-value definition,

$$d_k(\boldsymbol{\lambda}^{k+1}; \boldsymbol{z}^k) \leq \mathcal{P}_k\big(\boldsymbol{x}_k(\boldsymbol{\lambda}^{k+1}; \boldsymbol{z}^{k+1}), \boldsymbol{\lambda}^{k+1}; \boldsymbol{z}^k\big).$$

Subtracting this inequality from the corresponding equality at $\boldsymbol{z}^{k+1}$ and expanding the quadratic terms yields

$$\begin{aligned}
&d_k(\boldsymbol{\lambda}^{k+1}; \boldsymbol{z}^{k+1}) - d_k(\boldsymbol{\lambda}^{k+1}; \boldsymbol{z}^k) \\
&\quad \geq \frac{1}{\gamma} \left\langle \boldsymbol{z}^k - \boldsymbol{x}_k(\boldsymbol{\lambda}^{k+1}; \boldsymbol{z}^{k+1}), \Delta_{\boldsymbol{z}}^{k+1} \right\rangle + \frac{1}{2\gamma} \|\Delta_{\boldsymbol{z}}^{k+1}\|^2 \\
&\quad = -\frac{\alpha_k(2-\alpha_k)}{2\gamma} \|\boldsymbol{x}^{k+1} - \boldsymbol{z}^k\|^2 \\
&\qquad + \frac{\alpha_k}{\gamma} \left\langle \boldsymbol{x}^{k+1} - \boldsymbol{x}_k(\boldsymbol{\lambda}^{k+1}; \boldsymbol{z}^{k+1}), \boldsymbol{x}^{k+1} - \boldsymbol{z}^k \right\rangle.
\end{aligned}$$

Multiply this inequality by -2 and substitute it into the preceding estimate. Its squared-step term combines with the $\boldsymbol{z}$-step term in (6.13) to give $\alpha_k(2-\alpha_k)\|\boldsymbol{x}^{k+1} - \boldsymbol{z}^k\|^2/(2\gamma)$. Both parameter-drift terms remain unchanged, proving (6.15). ∎

Proof of Lemma 6.7. By Lemma 6.4, p_k has a $1/\gamma$-Lipschitz continuous gradient and

$$\nabla p_k(\boldsymbol{z}^k) = \frac{1}{\gamma}\big(\boldsymbol{z}^k - \boldsymbol{x}_k^\star(\boldsymbol{z}^k)\big).$$

The $\boldsymbol{z}$-update gives

$$\boldsymbol{z}^{k+1} - \boldsymbol{z}^k = \alpha_k(\boldsymbol{x}^{k+1} - \boldsymbol{z}^k).$$

Applying the smoothness inequality for p_k and expanding the inner product yields (6.16). ∎

Proof of Lemma 6.8. By Lemma 6.7,

$$\begin{aligned} 2\big[p_k(\boldsymbol{z}^{k+1}) - p_k(\boldsymbol{z}^k)\big] \le & -\frac{\alpha_k(2-\alpha_k)}{\gamma}\|\boldsymbol{x}^{k+1} - \boldsymbol{z}^k\|^2 \\ & + \frac{2\alpha_k}{\gamma}\left\langle \boldsymbol{x}^{k+1} - \boldsymbol{x}_k^\star(\boldsymbol{z}^k), \boldsymbol{x}^{k+1} - \boldsymbol{z}^k\right\rangle. \end{aligned} \tag{.26}$$

Adding this inequality to (6.15), the quadratic terms combine as

$$\frac{\alpha_k(2-\alpha_k)}{2\gamma} - \frac{\alpha_k(2-\alpha_k)}{\gamma} = -\frac{\alpha_k(2-\alpha_k)}{2\gamma}.$$

The two inner products combine into

$$\frac{2\alpha_k}{\gamma}\left\langle \boldsymbol{x}_k(\boldsymbol{\lambda}^{k+1}; \boldsymbol{z}^{k+1}) - \boldsymbol{x}_k^\star(\boldsymbol{z}^k), \boldsymbol{x}^{k+1} - \boldsymbol{z}^k\right\rangle.$$

Substitution into (6.15) proves (6.17). ∎

Proof of Lemma 6.9. The proximal optimality condition gives $\boldsymbol{q}(\boldsymbol{x}, \boldsymbol{\lambda}) \in \partial h(\boldsymbol{y}(\boldsymbol{x}, \boldsymbol{\lambda}))$, and hence $\|\boldsymbol{q}(\boldsymbol{x}, \boldsymbol{\lambda})\| \le L_h$. Substituting $\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}(\boldsymbol{x}, \boldsymbol{\lambda}) = (\boldsymbol{q}(\boldsymbol{x}, \boldsymbol{\lambda}) - \boldsymbol{\lambda})/\rho$ into the dual update yields

$$\boldsymbol{\lambda}^{k+1} = \left[1 - \tau_k\left(\frac{1}{\rho} + \frac{1}{s_k}\right)\right]\boldsymbol{\lambda}^k + \frac{\tau_k}{\rho}\boldsymbol{q}(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k) + \frac{\tau_k}{s_k}\boldsymbol{\mu}^k.$$

The coefficients are nonnegative and sum to one. Together with $\boldsymbol{\mu}^{k+1} = (1-\beta_k)\boldsymbol{\mu}^k + \beta_k\boldsymbol{\lambda}^{k+1}$, this proves $\|\boldsymbol{\lambda}^k\|, \|\boldsymbol{\mu}^k\| \le M$ by induction from the definition of M. The center update then gives $\|\boldsymbol{\mu}^{k+1} - \boldsymbol{\mu}^k\| \le 2M\beta_k$.

At the saddle point, the dual first-order condition gives

$$\boldsymbol{\lambda}_k^\star(\boldsymbol{z}) = \frac{s_k}{\rho + s_k}\boldsymbol{q}\big(\boldsymbol{x}_k^\star(\boldsymbol{z}), \boldsymbol{\lambda}_k^\star(\boldsymbol{z})\big) + \frac{\rho}{\rho + s_k}\boldsymbol{\mu}^k.$$

Since both vectors in this convex combination have norm at most M, $\|\boldsymbol{\lambda}_k^\star(\boldsymbol{z})\| \le M$. ∎

Proof of Lemma 6.10. Fix k and $\boldsymbol{z}$. Strong convexity gives the unique minimizer $\boldsymbol{x}_k(\boldsymbol{\lambda}; \boldsymbol{z})$ for each $\boldsymbol{\lambda}$. The finite, smooth, $1/s_k$-strongly concave function $d_k(\cdot; \boldsymbol{z})$ tends to $-\infty$ as $\|\boldsymbol{\lambda}\| \to \infty$, so it has a unique maximizer. The minimax equality and strong convexity in $\boldsymbol{x}$ identify the unique saddle pair and give $\boldsymbol{x}_k^\star(\boldsymbol{z}) = \boldsymbol{x}_k(\boldsymbol{\lambda}_k^\star(\boldsymbol{z}); \boldsymbol{z})$.

The maximizing property at $\boldsymbol{x}_k^\star(\boldsymbol{z})$ and strong convexity at fixed $\boldsymbol{\lambda}$ imply

$$\begin{aligned} p_k(\boldsymbol{z}) - d_k(\boldsymbol{\lambda}; \boldsymbol{z}) &\ge \mathcal{P}_k(\boldsymbol{x}_k^\star(\boldsymbol{z}), \boldsymbol{\lambda}; \boldsymbol{z}) - \mathcal{P}_k(\boldsymbol{x}_k(\boldsymbol{\lambda}; \boldsymbol{z}), \boldsymbol{\lambda}; \boldsymbol{z}) \\ &\ge \frac{\mu_\gamma}{2}\|\boldsymbol{x}_k(\boldsymbol{\lambda}; \boldsymbol{z}) - \boldsymbol{x}_k^\star(\boldsymbol{z})\|^2. \end{aligned}$$

This proves (6.19). For the center sensitivity of $\boldsymbol{x}_k(\boldsymbol{\lambda};\boldsymbol{z})$, add the strong-convexity inequalities at the minimizers for $\boldsymbol{z}$ and $\boldsymbol{z}'$. The only center-dependent term is $\|\boldsymbol{x}-\boldsymbol{z}\|^2/(2\gamma)$, so

$$\mu_\gamma\|\boldsymbol{x}_k(\boldsymbol{\lambda};\boldsymbol{z}')-\boldsymbol{x}_k(\boldsymbol{\lambda};\boldsymbol{z})\|^2 \le \frac{1}{\gamma}\left\langle \boldsymbol{z}'-\boldsymbol{z}, \boldsymbol{x}_k(\boldsymbol{\lambda};\boldsymbol{z}')-\boldsymbol{x}_k(\boldsymbol{\lambda};\boldsymbol{z})\right\rangle.$$

For the saddle primal points, use the same argument for $\max_{\boldsymbol{\lambda}} \mathcal{P}_k(\cdot,\boldsymbol{\lambda};\boldsymbol{z})$, whose strong-convexity modulus is also μ_γ. Cauchy–Schwarz gives (6.20) in both cases. The latter argument also gives

$$\langle \boldsymbol{z}'-\boldsymbol{z}, \boldsymbol{x}_k^\star(\boldsymbol{z}')-\boldsymbol{x}_k^\star(\boldsymbol{z})\rangle \ge \gamma\mu_\gamma\|\boldsymbol{x}_k^\star(\boldsymbol{z}')-\boldsymbol{x}_k^\star(\boldsymbol{z})\|^2.$$

Together with the Danskin formula for ∇p_k and $2\gamma\mu_\gamma > 1$, this verifies the smoothness bound used above:

$$\begin{aligned}\gamma^2\|\nabla p_k(\boldsymbol{z}')-\nabla p_k(\boldsymbol{z})\|^2 &\le \|\boldsymbol{z}'-\boldsymbol{z}\|^2-(2\gamma\mu_\gamma-1)\|\boldsymbol{x}_k^\star(\boldsymbol{z}')-\boldsymbol{x}_k^\star(\boldsymbol{z})\|^2\\ &\le \|\boldsymbol{z}'-\boldsymbol{z}\|^2.\end{aligned}$$

□

Proof of Lemma 6.11. Let $u_{k+1} := (\boldsymbol{x}^{k+1},\boldsymbol{\lambda}^{k+1},\boldsymbol{z}^{k+1})$ and define

$$\widehat{\Psi}_{k+1|k} := \mathcal{P}_{k+1}(\boldsymbol{x}^{k+1},\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^{k+1}) - 2d_k(\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^{k+1}) + 2p_k(\boldsymbol{z}^{k+1}).$$

Then

$$\Psi_{k+1}(u_{k+1})-\Psi_k(u_k) = \widehat{\Psi}_{k+1|k}-\Psi_k(u_k)+R_k,$$

where

$$R_k := 2\big[d_k(\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^{k+1})-d_{k+1}(\boldsymbol{\lambda}^{k+1};\boldsymbol{z}^{k+1})\big]+2\big[p_{k+1}(\boldsymbol{z}^{k+1})-p_k(\boldsymbol{z}^{k+1})\big].$$

Apply Lemma 6.8 to the first difference. Decompose the indefinite term at $\boldsymbol{x}_k^\star(\boldsymbol{z}^{k+1})$ and apply Lemma 6.10, followed by Young's inequality. The four bounds in (6.21) absorb the resulting cross terms and leave

$$\begin{aligned}\widehat{\Psi}_{k+1|k}-\Psi_k(u_k) \le\ & -\frac{1}{4\eta_k}\|\Delta_{\boldsymbol{x}}^{k+1}\|^2-\frac{1}{8\tau_k}\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2-\frac{\alpha_k}{2\gamma}\|\boldsymbol{x}^{k+1}-\boldsymbol{z}^k\|^2\\ &+\frac{\beta_k(2-\beta_k)}{2s_k}\|\boldsymbol{\lambda}^{k+1}-\boldsymbol{\mu}^k\|^2\\ &+\frac{1}{2}\left(\frac{1}{s_k}-\frac{1}{s_{k+1}}\right)\|\boldsymbol{\lambda}^{k+1}-\boldsymbol{\mu}^{k+1}\|^2.\end{aligned}$$

It remains to bound R_k. Lemma 6.9 gives $\|\boldsymbol{\lambda}^j\|, \|\boldsymbol{\mu}^j\|, \|\boldsymbol{\lambda}_j^\star(\boldsymbol{z})\| \le M$ and $\|\boldsymbol{\mu}^{k+1}-\boldsymbol{\mu}^k\| \le 2M\beta_k$. Direct comparison of the two quadratic damping terms gives, uniformly in $\boldsymbol{x}$ and $\boldsymbol{z}$,

$$|d_{k+1}(\boldsymbol{\lambda};\boldsymbol{z})-d_k(\boldsymbol{\lambda};\boldsymbol{z})| \le \Delta_k, \qquad |p_{k+1}(\boldsymbol{z})-p_k(\boldsymbol{z})| \le \Delta_k,$$

where

$$\Delta_k := 2M^2\left(\frac{1}{s_k}-\frac{1}{s_{k+1}}\right)+\frac{2M}{s_k}\|\boldsymbol{\mu}^{k+1}-\boldsymbol{\mu}^k\|.$$

Consequently,

$$|R_k| \le 4\Delta_k \le 8M^2\left(\frac{1}{s_k}-\frac{1}{s_{k+1}}\right)+16M^2\frac{\beta_k}{s_k}.$$

The two explicit positive terms above satisfy

$$\frac{\beta_k(2-\beta_k)}{2s_k}\|\boldsymbol{\lambda}^{k+1}-\boldsymbol{\mu}^k\|^2 \le 4M^2\frac{\beta_k}{s_k},$$

and

$$\frac{1}{2}\left(\frac{1}{s_k}-\frac{1}{s_{k+1}}\right)\|\boldsymbol{\lambda}^{k+1}-\boldsymbol{\mu}^{k+1}\|^2 \le 2M^2\left(\frac{1}{s_k}-\frac{1}{s_{k+1}}\right).$$

Adding the estimates proves (6.23).

For the stated schedule, the two error series satisfy

$$\sum_{k=0}^{\infty}\left(\frac{1}{s_k}-\frac{1}{s_{k+1}}\right)=\frac{1}{c_s}, \qquad \sum_{k=0}^{\infty}\frac{\beta_k}{s_k}\le\frac{4\theta_\mu}{c_s},$$

which makes the accumulated transport error finite. The lower bound (6.22) follows as follows. For any $\boldsymbol{\lambda}\in\partial h(\mathcal{A}(\boldsymbol{x}))$, proximal optimality gives $\boldsymbol{y}(\boldsymbol{x},\boldsymbol{\lambda})=\mathcal{A}(\boldsymbol{x})$ and hence $\mathcal{D}_\rho(\boldsymbol{x},\boldsymbol{\lambda})=F(\boldsymbol{x})$. Since $\|\boldsymbol{\lambda}\|\le L_h$ and $\|\boldsymbol{\mu}^k\|\le M$,

$$\max_{\boldsymbol{\lambda}}\mathcal{P}_k(\boldsymbol{x},\boldsymbol{\lambda};\boldsymbol{z})\ge F(\boldsymbol{x})+\frac{1}{2\gamma}\|\boldsymbol{x}-\boldsymbol{z}\|^2-\frac{(L_h+M)^2}{2s_k}.$$

Minimizing in $\boldsymbol{x}$ and using $s_k\ge s_0$ proves (6.22), because $\Psi_k=(\mathcal{P}_k-d_k)+(p_k-d_k)+p_k\ge p_k$. Summing (6.23) and using the error budget then gives

$$\begin{aligned}&\sum_{k=0}^{N-1}\left[\frac{1}{4\eta_k}\|\Delta_{\boldsymbol{x}}^{k+1}\|^2+\frac{1}{8\tau_k}\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|^2+\frac{\alpha_k}{2\gamma}\|\boldsymbol{x}^{k+1}-\boldsymbol{z}^k\|^2\right]\\ &\quad\le\Psi_0(\boldsymbol{x}^0,\boldsymbol{\lambda}^0;\boldsymbol{z}^0)-\inf_{\boldsymbol{x}\in\mathbb{R}^n}F(\boldsymbol{x})+\frac{(L_h+M)^2}{2s_0}+\frac{(10+80\theta_\mu)M^2}{c_s}, \qquad N\ge 1.\end{aligned}$$

∎

Proof of Theorem 6.1. The schedule (6.2) satisfies (6.21): $\eta_k L_x=1/2$, $\tau_k L_{d,k}\le\tau_k L_{d,0}\le 1/2$, $\tau_k G_{\mathcal{A}}^2\le\mu_\gamma^2/(72L_x)=\eta_k\mu_\gamma^2/36$, and the two bounds on α_k follow from $s_k\ge s_0$ and its definition. Moreover, $s_{k+1}\ge s_k>0$, $\eta_k,\tau_k>0$, and $0<\beta_k\le 1$. Since $L_{d,k}\ge 1/\rho+1/s_k$, the multiplier-boundedness condition also holds. By assumption, $\inf_{\boldsymbol{x}}F(\boldsymbol{x})>-\infty$. The master budget proved after Lemma 6.11 is therefore finite and independent of K.

The tail has $\lfloor K/2\rfloor\ge K/3$ elements. Since the primal and dual stepsizes are fixed and $\alpha_k\ge\alpha_{K-1}=\Theta(K^{-1/3})$ on the tail, the master budget and the definition of κ give

$$\|\Delta_{\boldsymbol{x}}^{\kappa+1}\|^2=O(K^{-1}), \qquad \|\Delta_{\boldsymbol{\lambda}}^{\kappa+1}\|^2=O(K^{-1}), \qquad \|\boldsymbol{x}^{\kappa+1}-\boldsymbol{z}^{\kappa}\|^2=O(K^{-2/3}).$$

The output certificate follows directly from (2.4). For stationarity, the $\boldsymbol{x}$-update and $L_{\mathcal{D},\boldsymbol{x}}$-smoothness of $\mathcal{D}_\rho(\cdot,\boldsymbol{\lambda}^k)$ give

$$\|\nabla f(\boldsymbol{x}^{k+1})+J_{\mathcal{A}}(\boldsymbol{x}^{k+1})^\top\boldsymbol{q}^k\|\le\left(\frac{1}{\eta_k}+L_x\right)\|\Delta_{\boldsymbol{x}}^{k+1}\|+\frac{1}{\gamma}\|\boldsymbol{x}^{k+1}-\boldsymbol{z}^k\|.$$

For feasibility, the dual update gives

$$\mathcal{A}(\boldsymbol{x}^{k+1})-\boldsymbol{y}^{k+1}=\frac{1}{\tau_k}\Delta_{\boldsymbol{\lambda}}^{k+1}+\frac{1}{s_k}(\boldsymbol{\lambda}^k-\boldsymbol{\mu}^k),$$

and hence

$$\|\mathcal{A}(\boldsymbol{x}^{k+1})-\boldsymbol{y}^{k+1}\|\le\frac{1}{\tau_k}\|\Delta_{\boldsymbol{\lambda}}^{k+1}\|+\frac{2M}{s_k}.$$

Square the last two bounds using $(a+b)^2\le 2a^2+2b^2$ and use $s_\kappa^{-2}=O(K^{-2/3})$. Both squared residuals are $O(K^{-2/3})$, so their norms are $O(K^{-1/3})$. All constants depend only on the problem data and fixed initialization and parameters, so $K=O(\epsilon^{-3})$ suffices. ∎

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